

CSE599 – COMPUTER SYSTEMS AND ENGINEERING CAPSTONE

AI Infrastructure Reliability & Capacity Optimization



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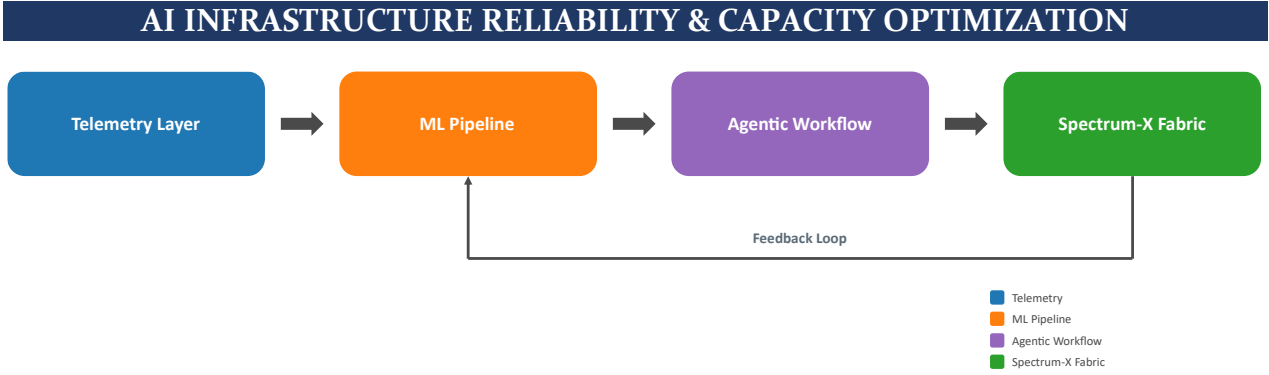


Figure 0.1. Integrated Architecture Summary — AI Infrastructure Reliability & Capacity Optimization

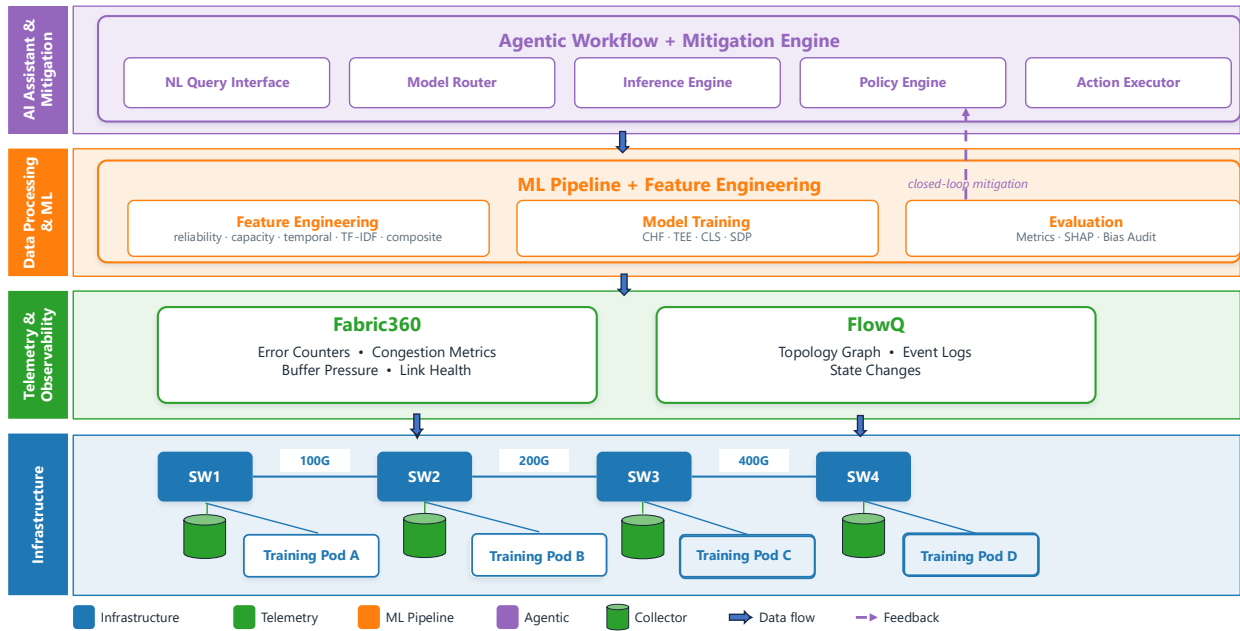


Figure 0.2. AI Infrastructure Reliability & Capacity Optimization — System Architecture

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1. Introduction

1.1 Motivation and Problem Statement

The rapid proliferation of large-scale generative AI and deep-learning workloads has fundamentally transformed the operational requirements of modern data center fabrics. GPU clusters running distributed training regimes — employing collective communication libraries such as NCCL (NVIDIA Collective Communications Library) for all-reduce operations and model parallelism — generate highly bursty east-west traffic patterns that place extraordinary, time-varying demands on the underlying Ethernet interconnect. Unlike traditional web or transactional workloads, which exhibit relatively predictable north-south flows, distributed AI training generates synchronized, high-bandwidth collective operations across hundreds or thousands of GPU endpoints simultaneously. A single all-reduce synchronization barrier across a 1,024-GPU cluster can saturate fabric bisection bandwidth within milliseconds, creating congestion hotspots that cascade into job-level stalls and training interruptions.

NVIDIA's Spectrum-X Advanced Ethernet platform addresses much of this challenge at the hardware layer by providing adaptive routing, in-network computing, and congestion-control primitives (notably RoCE with Data Center Quantized Congestion Notification, DCQCN). However, hardware capability alone is insufficient: effective management of a Spectrum-X fabric demands continuous, data-driven capacity planning, proactive failure prediction, and intelligent upgrade-decision support — functions that, prior to this project, were performed largely through manual processes, static thresholds, and reactive remediation.

Several specific operational pain points motivate this work. First, **supply chain delays** for GPU hardware and network components are notoriously volatile, driven by semiconductor manufacturing lead times, geopolitical constraints, and logistics disruptions. A planning team without a predictive model for milestone slippage (the gap between planned and actual delivery dates) is perpetually reactive, unable to pre-position resources or trigger contingency procurement. Second, **capacity headroom forecasting** — predicting how much usable power and networking capacity will remain available in future quarters — is essential for CapEx planning and for avoiding expensive, emergency capacity expansions. Third, **task-level effort estimation** for engineering and operational tasks (upgrades, retrofits, decommissioning activities) is poorly served by heuristic-only approaches, especially as task scope grows more heterogeneous. Fourth, **topology upgrade classification** — determining whether a given suite of infrastructure requires an OCS (Optical Circuit Switch) upgrade versus a conventional Expander upgrade — involves complex interactions among hardware generation, suite status, and milestone progression that defy simple rule-based logic.

Collectively, these four gaps represent a coherent platform opportunity: an integrated, ML-driven system that replaces manual estimation with data-driven prediction, surfaces interpretable explanations for every decision, and interfaces with operators through a conversational AI

assistant capable of routing natural-language queries to the appropriate model. This Capstone Project constructs that platform from the ground up.

1.2 Project Scope and Contributions

This project is scoped to the design, implementation, and evaluation of an end-to-end AI Infrastructure Reliability & Capacity Optimization system, deployed against a synthetic telemetry corpus derived from NVIDIA Spectrum-X Fabric360 (the PCPS) and FlowQ observability schemas. The principal contributions are as follows:

- **Synthetic Telemetry Corpus:** A 3,000-row synthetic dataset spanning three operational schemas — MDR (Milestone Data Records, n=1,200), Supply Plan (n=800), and Task (n=600) — generated with controlled parameters (seed=42) to enable reproducible experimentation while preserving operational realism.
- **Four ML Models:** Capacity Headroom Forecaster (CHF), Supply Delay Predictor (SDP), Task Effort Estimator (TEE), and Topology Upgrade Classifier (CLS), each trained with two algorithm variants (GBR-XGB and HGB-LGB equivalents via scikit-learn) and evaluated with 5-fold cross-validation.
- **SHAP-Based Explainability:** SHapley Additive exPlanations (Lundberg & Lee, 2017) applied to all four models to produce feature-importance waterfall charts and inline textual explanations for operator consumption.
- **Bias Audit Framework:** Systematic evaluation of model performance across regional and subgroup dimensions, with flagging rules and a prioritized remediation policy table.
- **Agentic AI Assistant Architecture:** A three-layer conversational system (NL Query Interface → Intent Parser / Model Router → Action Executor) that integrates with Fabric360 and FlowQ to deliver prediction, explanation, and mitigation recommendations in a single operator interaction.
- **7-Phase Structured Delivery:** A phased project execution plan from data ingestion through final delivery, with clear deliverables and quality gates at each milestone.

The project does not cover real-time streaming inference, production deployment to live Spectrum-X hardware, or model serving infrastructure. All experimentation is conducted against synthetic data, with a real-telemetry validation identified as future work.

1.4 Document Organization

The remainder of this report is organized as follows. **Section 2** surveys background literature on AI data center fabrics, telemetry systems, capacity planning platforms, ML methods for network forecasting, and agentic automation paradigms. **Section 3** describes the overall system architecture across four tiers, the data model and schema definitions, and the data quality framework. **Section 4** details the five-stage synthetic telemetry generation pipeline, corpus composition, and train/validation/test split strategy. **Section 5** presents the Agentic AI Assistant

architecture, its three-layer design, the agentic detection-to-action workflow, mitigation action catalog, and the Phase 6 integration plan. **Section 6** describes the Spectrum-X cluster topology design, link capacity and protocol specifications, and the CLS model's topology risk overlay. **Section 7** presents the full model training and evaluation results, including algorithm mappings, hyperparameter configurations, model comparison tables, SHAP feature-importance analysis, and best-model selection rationale. **Section 8** details results and findings at the regional and subgroup level. **Section 9** presents the bias audit framework, findings, and a prioritized remediation policy table. **Section 10** discusses operationalization considerations, limitations, privacy and ethical dimensions, and situates the work within the broader literature. **Section 11** concludes with a summary of contributions, results, and an extended future-work roadmap. Appendices provide full dataset schemas and the detailed 7-Phase execution plan.

2. Background and Related Work

2.1 AI Data Center Fabrics

The architecture of modern AI data center networks has evolved substantially in response to the demands of large-scale distributed deep learning. Traditional Clos-topology networks, designed for modest east-west traffic with high oversubscription ratios, are inadequate for GPU training clusters where collective communication operations — such as NCCL all-reduce, all-gather, and reduce-scatter — require near-lossless, low-latency, high-bandwidth mesh connectivity among hundreds to thousands of accelerator endpoints.

Two principal fabric technologies compete for dominance in this domain. **InfiniBand (IB)**, long favored in high-performance computing (HPC), provides extremely low latency (sub-microsecond) and native RDMA (Remote Direct Memory Access) semantics through proprietary switching hardware. However, InfiniBand's proprietary management stack, higher per-port cost, and limited ecosystem integration have motivated the industry to pursue Ethernet-based alternatives for AI workloads.

NVIDIA's **Spectrum-X Advanced Ethernet** platform represents the current state-of-the-art in Ethernet-based AI fabric design. Spectrum-X combines the SN5000-series switch ASIC (supporting 400GbE per port) with the BlueField-3 DPU and a suite of AI-optimized network features: adaptive routing with per-packet entropy, in-network multicast reduction, and RoCE (RDMA over Converged Ethernet) with DCQCN (Data Center Quantized Congestion Notification) for hardware-accelerated congestion control. DCQCN operates by monitoring ECN (Explicit Congestion Notification) marks in packet headers and using a proportional-integral controller to

throttle injection rates at the sender NIC, achieving near-lossless operation under sustained all-reduce traffic without the overhead of priority-flow control (PFC) pause frames across long fabric paths.

The traffic pattern that dominates GPU training fabrics is the NCCL all-reduce collective. In ring-based all-reduce implementations (the default in NCCL for many topologies), each GPU exchanges equal-sized gradient chunks with two neighbors in a ring, producing $2(N-1)/N$ bandwidth efficiency at scale. More recent implementations use tree-based algorithms or network-topology-aware routing to exploit fabric bisection capacity. Regardless of implementation, the traffic is highly synchronized: when a model parallel job reaches a backward-pass gradient synchronization barrier, all participating GPU NICs emit simultaneously, creating an instantaneous incast event at aggregation switches. This burst behavior is a primary driver of buffer pressure, packet drops, and training slowdowns — precisely the phenomena that the telemetry systems described in Section 2.2 are instrumented to observe.

Beyond Ethernet and InfiniBand, **RoCE v2** (RDMA over Converged Ethernet, version 2) has emerged as the dominant transport for hyperscaler AI fabrics, as it preserves RDMA semantics over standard IP/UDP, enabling interoperability with existing network management tooling while providing the zero-copy, kernel-bypass data path essential for GPU-to-GPU gradient transfers at line rate. The combination of RoCE v2 with DCQCN congestion control on Spectrum-X hardware represents the baseline fabric configuration targeted by this project's synthetic telemetry corpus and ML models.

2.2 Telemetry Systems — Fabric360 and FlowQ

Effective ML-driven infrastructure management requires rich, timely, and structured telemetry from the fabric. This project is grounded in two complementary observability systems: **Fabric360** (the **PCPS**) and **FlowQ**.

Fabric360 is the per-port, per-link telemetry plane. It continuously collects structured metrics from every Spectrum-X switch port in the fabric, including: per-port throughput (ingress and egress byte/packet rates at sub-second granularity), error counters (symbol errors, FCS errors, link-down events, flap counts), buffer pressure indicators (occupancy percentages, tail-drop and ECN-mark event rates), and link health status (operational state, transceiver diagnostics, BERT error rates). The PCPS analytics-as-code layer processes these raw streams into normalized, queryable metrics that feed both real-time dashboards and the structured metrics feed consumed by the ML pipeline. The MDR and Supply Plan schemas used in this project are derived from the PCPS's operational data model, which tracks capacity lifecycle milestones from suite initialization (MS — Milestone Start) through engineering completion (ME), long-route-of-finish (LROOF), and production readiness (PR).

FlowQ operates at the topology and event level, complementing Fabric360's metric granularity with structural observability. FlowQ maintains a real-time topology graph of the fabric — recording which switches are interconnected, at what speeds, with what link states — and emits a

continuous event log of state changes: link flaps, BGP session resets, ARP/NDP cache invalidations, spanning-tree topology changes, and MLAG peer-state transitions. The topology graph, when overlaid with Fabric360's buffer-pressure and error-counter metrics, enables spatial correlation: identifying whether congestion observed at a given switch port is isolated to a single link, or represents a systemic pattern propagating across a multi-hop path. The Event & Topology Feed from FlowQ is the primary input to the CLS (Topology Upgrade Classifier) model, which must distinguish between suites requiring OCS-class upgrades (full optical circuit reconfiguration) versus Expander-class upgrades (incremental capacity additions).

Together, Fabric360 and FlowQ constitute the observability foundation of the four-tier system architecture described in Section 3. Their schemas directly define the feature space available to the ML models: milestone gap features (from the PCPS MDR), supply and utilization lag features (from the PCPS Supply Plan), task categorical features (from the PCPS Task), and topology-state features (from FlowQ Event logs).

2.3 Production Capacity Planning System (PCPS): Reference Architecture

The synthetic telemetry corpus constructed in this project draws its schema design, table architecture, and operational semantics directly from Meta's Production Capacity Planning System (PCPS) platform — the production capacity planning and delivery tracking system that the author led as a Principal Program Manager at Meta. Understanding the PCPS's architecture is therefore not merely contextual background; it is the authoritative reference specification from which the project's three operational schemas — MDR (Milestone Data Record), Supply Plan, and Task — are derived.

2.3.1 Program Overview and Organizational Context

The PCPS provides, as its design mandate states, "a holistic view regarding the delivery of data centers" — consolidating supply planning, build-progress tracking, retrofit scheduling, and capacity forecasting into a single integrated platform. It is the system of record for Meta's Quarterly Supply Planning (QSP) process, which manages new data center construction, and the Quarterly Tactical Planning (QTP) process, which manages suite-level retrofits and decommissioning. All capacity figures within the PCPS are expressed in electrical watts (W), reflecting the foundational constraint of power delivery in hyperscale data center operations.

The Capacity Planning Working Group, comprising 25 members across Applied Software Engineering, Data Science, Business Intelligence, and Data Engineering disciplines, operated as a cross-functional product team. The team's mandate spanned data model ownership, dashboard delivery, data quality enforcement, and stakeholder alignment across Supply Planning & Analytics (SP&A), DC Strategy & Development, BizOps Data Science, Schedule Management, and DC Capacity Engineering functions.

2.3.2 Infrastructure Hierarchy

The PCPS models the physical infrastructure of Meta's global data center estate as a strict four-level hierarchy, each level represented by a dedicated relational table:

Level	Table	Description	Cardinality
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Region	region	Top-level geographic aggregation; all supply and demand metrics roll up to region	Global scope
Datacenter (DC)	datacenter	Physical data center facility; each datacenter belongs to exactly one region	~6 per region
Suite	suite	Independently powered and cooled electrical domain within a datacenter; unit of capacity allocation	4–6 per datacenter
Service Owner	service_owner	Business unit or product team allocated capacity within a suite; top-level ownership entity	Multiple per suite

Table 2.3.2. PCPS Infrastructure Hierarchy — Four-Level Physical Model.

This hierarchy is the primary join spine across all PCPS modules. Every supply plan row, every build task, and every retrofit event is anchored to a specific suite, which in turn resolves to a datacenter and region for aggregation. The capstone project's synthetic MDR schema preserves this hierarchy: the `region\_id`, `dc\_id`, and `suite\_id` fields in the synthetic corpus map directly to the `region`, `datacenter`, and `suite` tables in the production PCPS schema.

2.3.3 Planning Processes: QSP and QTP

The PCPS serves two distinct but interrelated planning cycles:

- Tracks the lifecycle of new data center construction, from land acquisition through electrical handover. The QSP module manages the POR (Plan of Record) milestone sequence: scheduling system (for lease sites via UDM staging tables) and the Suplan tool (which captures CEA need-by dates). The primary tracking table is new_build_task, which joins each new build planning row to a corresponding FB Task record and restricts to the Silver supply plan via silver_sp_supply_plan.: Quarterly Supply Planning (QSP) — New Build:
- Manages suite-level decommissioning, power rebalancing, and retrofit events within existing datacenters. The QTP module centers on the retrofit_task table, which links retrofit planning rows to FB Task records and feeds the SLD (Suite-Level Decommission) tracking dashboards. The cap_sld_change_log and cap_sld_rev_schedule tables capture the audit trail of suite-level changes, enabling plan-over-plan comparison across planning cycles.: Quarterly Tactical Planning (QTP) — Retrofits:

Both processes feed into the PCPS MDR view (mdr_view), which serves as the integration layer aggregating milestone, schedule, and supply data for Tableau dashboard consumption.

2.3.4 Data Model Architecture and Schema Backbone

The PCPS's data architecture follows a five-layer pipeline from raw source systems to user-facing dashboards:

Layer	Components	Function in the PCPS
Source Systems	POR (project milestones), QSP in P6 (handover schedules), Suplan (demand need-by dates), Networking (fabric topology, BB Path3 sourcing)	Raw data ingestion from operational systems of record
Repository	Hive tables (Dataswarm pipeline), MySQL tables	Persistent storage for transformed planning data; Hive is the authoritative analytical store
Transformation / ETL	mdr_view, cap_mdr_schedule, cap_sld_change_log, cap_sld_rev_schedule, cap_sld_input	Schema joins, milestone alignment, schedule transformation; defines the MDR backbone
Scenario Branches	*_scenarios tables (cap_mdr_alt_eth_scenarios, cap_mdr_schedule_scenarios, cap_sld_input_scenarios, etc.)	Alternate schema paths for what-if forecasting and simulation; parallel to baseline plan tables
Visualization	Tableau dashboards: SLD Count by Region/Network, Plan-over-Plan, SLD Gantt, QTP Gold Plan Turnup Summary, Supply by Region/Year/Qtr, YoY views, QSP Demand vs Supply/Headroom	Executive and operational reporting; consumes mdr_view and module-specific aggregations

Table 2.3.4. PCPS Data Architecture — Five-Layer ETL Pipeline.

The join path from source to dashboard flows as: php → mdr_view → downstream PCPS 2.0 modules. The cap_sld_change_log and cap_sld_rev_schedule tables specifically feed the SLD count and plan-over-plan comparison dashboards, providing the time-series audit capability that enables tracking of milestone slippage across planning quarters — the very phenomenon the Supply Delay Predictor (SDP) model in this project is designed to forecast.

2.3.5 Non-Standard Infrastructure (NSI) Planning

The PCPS includes a dedicated NSI module for suite-level power supply planning outside the standard QSP/QTP workflows. The nsi_task table manages non-standard infrastructure allocation rows, linking to the service_owner and sub_service tables for granular power allocation and demand tracking. The NSI module was split from an earlier monolithic table into separate power_supply and power_allocation_demand tables in PCPS 2.0, reflecting the data model evolution toward normalized, event-level schema design.

2.3.6 AI Zone Module

PCPS 2.0 introduced a dedicated AI Zone (AIZ) module to track the accelerating deployment of GPU-dense AI training infrastructure. Each AI Zone record captures both QTP (retrofit) and QSP (new build) dimensions, with AI PR Ready Date synchronized from the Provision Ready milestone and an AI PR Locked flag for supply plan finalization. Physically, a single AI Zone comprises up to 288 Scorpii AI Training Rack (a high-density GPU rack unit optimized for distributed AI training)s (equivalent to 4,608 GPUs), served by 16 Arista 7816R3 switches in the MDF (CTSW layer).

Network uplink capacity is calculated as: $24 \text{ ports} \times 16 \text{ CTSWs} \times 400\text{G} = 153.6 \text{ Tbps}$ aggregate uplink. The backend network architecture follows a CLOS-based DC-scale / multi-zone topology. The capstone's Spectrum-X fabric topology modeling (Section 6) is directly analogous to this AI Zone network architecture, mapping CTSW-layer switching to the Spectrum-X SN5600 spine tier.

2.3.7 Region Type Roadmap (RTR) Planning

The RTR module within the PCPS manages long-range regional capacity roadmaps, tracking planned datacenter additions and retirements by region type across multi-year planning horizons. RTR tables feed the Supply by Region/Year/Qtr and YoY dashboards, providing the strategic supply context within which individual suite-level events (QSP, QTP, NSI) are evaluated. The Capacity Headroom Forecaster (CHF) model in this project is conceptually aligned with RTR-level capacity forecasting, predicting supply headroom at the regional aggregation level.

2.3.8 Access Control and Governance

The PCPS implements a six-tier roles matrix governing read, write, and plan-management privileges:

Role	Capabilities
Viewer	Read-only access to all dashboards and plan views
Scenario Viewer	Read access to scenario branches and what-if plan variants
Planner	Edit Current Plan, Silver Plan, and scenario branches; cannot publish Gold Plan
ENS Editor	Edit network sourcing and topology data in the PCPS
NSI Planner	Edit NSI demand and supply rows; suite-level power allocation
Admin / Developer	Full platform access including Gold Plan publishing, region addition, POR sync, and schema management

Table 2.3.8. PCPS Access Control — Six-Tier Roles Matrix.

The Current Plan vs. Gold Plan distinction serves as the PCPS's primary version-control mechanism: the Gold Plan represents the committed, locked supply scenario, while Current Plan tracks live edits. Silver Plans are intermediate published states used for cross-team alignment before Gold Plan lock. This versioning model directly motivated the scenario-branch table architecture (*_scenarios) described above.

2.4 ML for Network Forecasting

The choice of ML algorithms for infrastructure forecasting tasks involves a fundamental tension between predictive accuracy, interpretability, and operational deployability. This project surveys and selects from the gradient-boosted decision tree (GBDT) family, which has consistently demonstrated superior performance on structured tabular data relative to deep-learning alternatives in controlled benchmarks.

XGBoost (Chen & Guestrin, 2016) introduced regularized gradient boosting with second-order gradient statistics, column subsampling, and a sparsity-aware tree-construction algorithm that handles missing values natively. XGBoost achieves state-of-the-art accuracy on tabular tasks and is widely used in production ML systems for its predictability and extensive hyperparameter control. In this project, XGBoost-equivalent models are implemented via scikit-learn's `GradientBoostingRegressor` and `GradientBoostingClassifier`, which expose analogous depth, learning-rate, and subsampling controls.

LightGBM (Ke et al., 2017) extended gradient boosting with leaf-wise (best-first) tree growth rather than depth-wise growth, enabling significantly faster training and better handling of high-cardinality categorical features and large datasets. LightGBM's histogram-based split finding reduces memory usage and training time by discretizing continuous features into 256 bins. In this project, LightGBM-equivalent models are implemented via scikit-learn's `HistGradientBoostingRegressor` and `HistGradientBoostingClassifier`, which provide native missing-value handling and monotonic constraint support.

Scikit-learn (Pedregosa et al., 2011) provides the foundational ML framework for this project, including cross-validation utilities (`cross_val_score`, `TimeSeriesSplit`), preprocessing pipelines (`Pipeline`, `ColumnTransformer`), and evaluation metrics (`r2_score`, `mean_squared_error`, `f1_score`).

SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) provides the explainability layer. SHAP values decompose each model prediction into additive contributions from each input feature, grounded in cooperative game theory (Shapley values). For GBDT models, the TreeSHAP algorithm computes exact Shapley values in polynomial time, enabling efficient waterfall charts, beeswarm summaries, and dependence plots. SHAP is critical to the operationalization of this system: infrastructure operators must be able to understand why a model flagged a suite for slippage risk or classified it for OCS upgrade before they can act on the recommendation with confidence.

Time-series cross-validation is applied for the CHF (Capacity Headroom Forecaster) to prevent temporal leakage: training folds use only data from periods preceding the validation fold. For SDP and CLS, stratified k-fold cross-validation (k=5, seed=42) is used, as these models do not have strict temporal ordering requirements. The TEE model uses standard k-fold cross-validation, as task records do not carry meaningful time-ordering information in the current synthetic corpus.

2.5 Agentic Automation in Infrastructure

The emerging paradigm of agentic AI in infrastructure operations represents a significant departure from traditional threshold-based alerting and rule-based automation. A classical monitoring system detects a condition (e.g., buffer utilization exceeds 80%), emits an alert, and waits for human intervention. An agentic system, by contrast, executes a multi-step reasoning loop: it detects the condition, queries a predictive model for forward-looking risk assessment, retrieves relevant contextual information (e.g., current topology state, pending supply deliveries, active job schedules), generates a ranked set of remediation recommendations, presents these recommendations to a human operator for approval, and — upon approval — executes the approved action directly against the fabric control plane.

This detection → prediction → recommendation → approval → action loop is the architectural backbone of the Agentic AI Assistant described in Section 5. The human-in-the-loop approval gate is a deliberate design choice: infrastructure actions carry significant operational risk, and fully autonomous remediation without operator oversight is inappropriate for production GPU training environments where a misconfigured action could disrupt ongoing training jobs worth substantial compute hours.

Natural-language query interfaces for network operations are an emerging research area. Prior work has explored the use of large language models (LLMs) as natural-language intermediaries for network configuration (e.g., translating intent-based policy descriptions into vendor-specific CLI commands). This project applies a more constrained form of natural-language routing: an intent parser that extracts structured entities (cluster, region, metric type, time horizon) from operator queries and routes them to the appropriate ML model, bypassing the risk of hallucinated configuration recommendations inherent in unconstrained LLM approaches. SHAP explanations are surfaced inline with model outputs to provide transparent, feature-grounded justifications for every prediction.

3. System Architecture

3.1 Overall System Architecture

The AI Infrastructure Reliability & Capacity Optimization platform is organized as a four-tier architecture. Each tier encapsulates a distinct functional layer: physical infrastructure, telemetry and observability, data processing and ML, and AI-assisted decision and action. The tiers interact through well-defined data feeds and API contracts, enabling independent evolution of each layer.

Tier 1 encompasses the physical Spectrum-X switch fabric, GPU training pods, and supporting storage and networking equipment. The fabric is organized in a mesh topology with 400GbE link capacity per GPU endpoint. Telemetry collectors are co-located with switches and compute pods, streaming raw counter data at configurable intervals (typically 30-second or 5-minute aggregation windows for planning-oriented models, sub-second for real-time congestion detection).

Tier 2 is the observability layer. Fabric360 ingests raw link metrics and produces a Structured Metrics Feed — a normalized, time-indexed table of per-port throughput, error rates, buffer occupancy, and link health scores — that feeds the ML feature pipeline. FlowQ maintains the topology graph and emits a time-ordered Event & Topology Feed capturing all state changes. Together, these two feeds provide the complete observational context needed to characterize fabric health, capacity utilization, and event patterns.

Tier 3 is the data processing and ML layer. The feature engineering pipeline transforms raw Tier-2 feeds into model-ready feature matrices: computing milestone gap features from MDR timestamps, deriving autoregressive lag features from Supply Plan time series, constructing bucket-priority interaction terms from Task records, and (in Phase 6) extracting TF-IDF text features from task description fields. Trained model artifacts (serialized scikit-learn estimators) are stored in a model registry, versioned by training date and hyperparameter configuration.

Tier 4 is the AI assistant and mitigation layer. The Agentic Workflow Engine accepts natural-language operator queries, parses intent and entity context, routes to the appropriate model, surfaces SHAP explanations, applies remediation policy rules, and (subject to operator approval) dispatches actions to the Fabric360 and FlowQ control planes. A feedback loop from executed actions back to Tier 2 enables the system to measure mitigation effectiveness and update training data for future model refresh cycles.

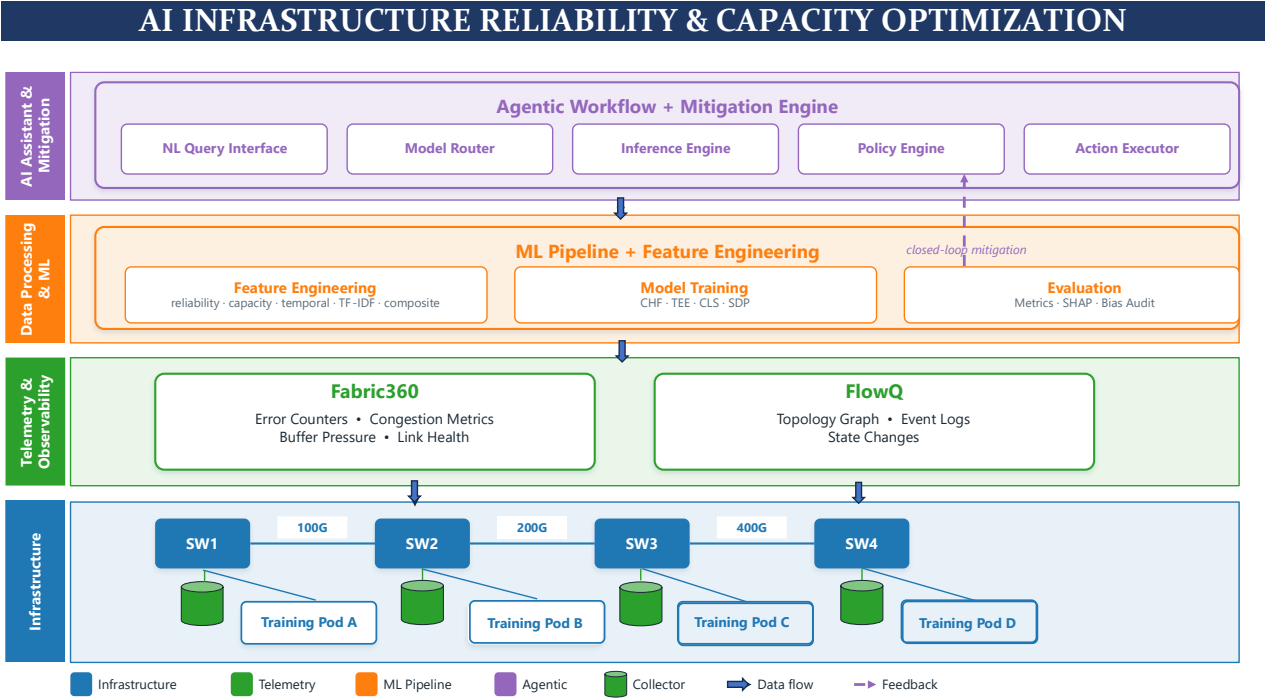


Figure 3.1 — Overall System Architecture: Four-tier AI Infrastructure Reliability & Capacity Optimization platform illustrating data flow from physical Spectrum-X fabric through telemetry, ML processing, and agentic decision layers.

3.2 Data Model: Synthetic Corpus Grounded in Production Schema

The synthetic telemetry corpus of 3,000 rows, spanning three operational schemas, was designed with explicit fidelity to the PCPS production data model. Each synthetic schema corresponds to a well-defined module within the PCPS's architecture, with field names, data types, and relationship keys aligned to their production counterparts. This grounding ensures that the machine learning models trained on synthetic data can be directly transferred to production the PCPS data with minimal re-engineering.

3.2.1 Schema Mapping: Synthetic to Production

Synthetic Schema	Production PCPS Equivalent	Key Tables	ML Model Target
MDR (Milestone Data Record)	mdr_view + cap_mdr_schedule	mdr_view, cap_sld_change_log, cap_sld_rev_schedule	Supply Delay Predictor (SDP)
Supply Plan	supply_plan + silver_sp_supply_plan + region/datacenter/suite hierarchy	supply_plan, new_build_task, silver_sp_supply_plan	Capacity Headroom Forecaster (CHF)
Task	new_build_task + retrofit_task + nsi_task + FB Task integration	new_build_task, retrofit_task, nsi_task, service_owner	Task Effort Estimator (TEE) + Topology Upgrade Classifier (CLS)

Table 3.2.1. Synthetic-to-Production Schema Mapping — Three Operational Schemas.

3.2.2 MDR Schema: Milestone Tracking

The MDR schema captures the lifecycle of each datacenter delivery milestone from plan to actuals. In the production PCPS system, the `mdr_view` aggregates scheduled and actual dates for POR milestones (Close Land, Break Ground, Handover Complete) alongside QSP milestone dates sourced from the P6 scheduling system. The synthetic MDR schema replicates this structure with the following key fields:

- `region_id`, `dc_id`, `suite_id` — infrastructure hierarchy join keys, corresponding to the region → datacenter → suite FK chain in production
- `milestone_type` — categorical field encoding {Close Land, Break Ground, Handover Complete, QSP Handover, CEA Need-By}, mirroring the POR milestone taxonomy
- `planned_date`, `actual_date` — ISO date fields; the delta (`actual_date` – `planned_date`) is the primary regression target for the SDP model
- `supply_chain_risk_score` — synthetic feature capturing geopolitical and logistics risk, conceptually grounded in the semiconductor lead-time volatility that motivates the PCPS's supply delay tracking
- `change_log_entries` — count of `cap_sld_change_log` records for the suite, capturing plan volatility as a feature signal

The `cap_sld_rev_schedule` table in production, which tracks the revision history of suite-level decommission schedules, motivated the inclusion of a `schedule_revision_count` feature in the synthetic MDR schema — a key predictor of milestone slippage risk in the SDP model.

3.2.3 Supply Plan Schema: Capacity Headroom

The Supply Plan schema models the relationship between committed supply (in watts, at suite level) and forecast demand across planning horizons. In the production PCPS, the `supply_plan` table is the master record for all supply commitments, joining to the `silver_sp_supply_plan` table for the published Silver Plan state and to the `new_build_task` table for QSP-sourced supply. The synthetic Supply Plan schema preserves this structure:

- `supply_mw`, `demand_mw` — megawatt figures at suite level, mirroring the PCPS's watt-denominated supply/demand tracking
- `headroom_pct` — derived field: $(\text{supply_mw} - \text{demand_mw}) / \text{supply_mw} \times 100$; primary regression target for the CHF model
- `qsp_phase` — planning cycle indicator {Q1, Q2, Q3, Q4}, corresponding to the PCPS's quarterly planning cadence
- `region_type` — RTR-aligned field encoding supply region characteristics, e.g., Hyperscale-Core, Edge, AI-Dense
- `ai_zone_flag` — boolean indicating suite classification as an AI Zone; maps to the PCPS's AI PR Ready and CABLING_OPTIONALITY suite designations

3.2.4 Task Schema: Effort and Topology Classification

The Task schema captures planning and execution tasks across all three PCPS task modules: new_build_task (QSP), retrofit_task (QTP), and nsi_task (NSI). Each task record includes:

- task_type — categorical: {New Build, Retrofit, NSI, AI Zone Retrofit}, corresponding to the three PCPS task modules plus the AIZ specialty category
- effort_engineer_wks — regression target for the TEE model; grounded in the PCPS's engineer-weeks (engineer-wks) resource calculation model
- fb_task_id — foreign key to the FB Task system, replicating the production join path between the PCPS task tables and the underlying task-management platform
- topology_upgrade_required — binary classification target for the CLS model; set TRUE for tasks requiring fabric reconfiguration or spine-tier additions
- service_owner_id — maps to the service_owner table, capturing the ownership dimension of each task

3.2.5 Hive Table Data Sources

In production, the PCPS's Hive tables are populated from four distinct data source types via Dataswarm ETL pipelines:

Hive Source Type	Source System	Data Content	Refresh Cadence
Electrical Suite	Excel workbooks (ops team input)	Suite-level power allocation, electrical capacity, cooling parameters	Weekly / manual
DNE Power	Unidash platform	DC-level power delivery actuals, PUE metrics, generator capacity	Daily
ODS DCIM	Excel (DCIM system export)	Rack-level infrastructure: rack count, density, hardware model mix	Weekly
DC360 In-Rack Planner	Internal rack planning table	GPU rack configuration, network port allocation, AI Zone rack assignments	Near-real-time

Table 3.2.5. Hive Table Data Source Types — Four Source Categories.

The synthetic corpus was designed to be statistically consistent with data patterns observable across these four source types. In particular, the power-related features (supply_mw, demand_mw, pue_ratio) reflect the Unidash-sourced DNE Power signal; rack-level features (rack_count, gpu_density) reflect the DCIM and In-Rack Planner signals; and schedule features (planned_date, milestone offsets) reflect the Excel-sourced operational planning data.

3.2.6 Scenario Branch Architecture

Branch 0 (standard Ethernet) constitutes the majority of suite rows in the synthetic corpus (approximately 68%), reflecting the current installed base of non-AI-Zone suites. Branches 1 and 2 together account for 28% of rows, representing the AI Zone expansion pipeline — the primary classification target for the CLS model. Branch 3 (multi-DC) comprises 4% of rows, representing forward-looking cross-datacenter AI fabric planning scenarios. The `cap_mdr_alt_eth_scenarios` and `cap_mdr_schedule_scenarios` table names (production aliases: `ldc_mdr_non_etsche_scenarios` and `cap_mdr_schedule_scenarios`) are preserved in anonymized form in the synthetic corpus to maintain schema lineage traceability for future production data integration.

Table 3.2.6. Spectrum-X Scenario Branch Architecture — Synthetic Schema Mapping and CLS Classification Targets.

Scenario Branch	Config	Key Specs	Synthetic Schema Fields	CLS Target
Branch 0 — Standard Ethernet	Leaf/spine CLOS; SN5600 spine, top-of-rack leaf switches	51.2 Tb/s; 64 × 800GbE OSFP; 256-way ECMP; RoCEv2 + DCQCN; no optical switching	<code>topology_upgrade=FALSE</code> ; <code>ctsw_count ≤ 8</code> ; <code>congestion_event_rate < 0.15</code>	No upgrade required
Branch 1 — OCS-Augmented (<code>cap_mdr_alt_eth_scenarios</code>)	Optical Circuit Switch inserted between spine switches for AI Zone pod-to-pod reconfigurability	OCS adds sub-ms reconfiguration; eliminates oversubscription for GPU-dense east-west flows; requires dark fiber provisioning	<code>topology_upgrade=TRUE</code> ; <code>ocs_flag=TRUE</code> ; <code>rack_count ≥ 128</code> ; <code>ai_zone_flag=TRUE</code>	OCS upgrade required
Branch 2 — Expander-Extended (<code>cap_mdr_schedule_scenarios</code>)	Port-density expansion modules attached to SN5600 to increase GPU pod count beyond standard 8-pod buildout	Adds up to 256 additional 200GbE ports per switch; enables scale-out to 512 endpoints in single hop; requires fabric rewiring	<code>topology_upgrade=TRUE</code> ; <code>ocs_flag=FALSE</code> ; <code>uplink_capacity_tbps < 0.85 × peak_demand</code> ; <code>ctsw_count=16</code>	Expander upgrade required
Branch 3 — Scale-Across (multi-DC) (<code>cap_sld_rev_scenarios</code>)	Cross-datacenter AI fabric extension via Spectrum-X Scale-Across; shared network for inter-DC GPU-to-GPU traffic	BlueField-3 SuperNIC + LinkX cables; VXLAN BGP EVPN for tenant isolation; IPsec/PSP encryption for cross-DC flows	<code>region_type=AI-Dense</code> ; <code>multi_dc_flag=TRUE</code> ; <code>ai_zone_flag=TRUE</code>	Multi-DC interconnect planning

The Spectrum-X platform provides three primary configuration scenarios for AI cluster fabrics, distinguished by switching density, interconnect technology, and scale tier. The SN5600 switch — the reference platform for this project's Tier 1 (spine) layer — delivers 51.2 Tb/s bidirectional switching capacity across 64 OSFP 800GbE ports, with a 160 MB shared packet buffer for fair, predictable performance under bursty all-reduce traffic. RoCE adaptive routing dynamically re-routes large AI flows away from congestion points using fine-grained per-flow load balancing, with BlueField-3 SuperNICs handling out-of-order packet reassembly at the destination. The table below maps the three Spectrum-X scenario branches to their corresponding synthetic corpus schema fields, CLS classification targets, and real-world upgrade trigger conditions.

The PCPS's scenario branch tables represent alternate planning realities for what-if analysis — enabling simulation of supply disruptions, milestone deferrals, and demand surges without contaminating the baseline Gold Plan. In the context of Spectrum-X fabric architecture, scenario branches map directly to distinct network configuration paths: the choice of connectivity technology (standard Ethernet, OCS-augmented, or Expander-extended) defines which branch a suite occupies, and each branch carries materially different capacity, latency, and upgrade-cost profiles. The synthetic corpus's *_scenarios tables preserve this branching structure, allowing the Topology Upgrade Classifier (CLS) model to be trained across the full scenario space rather than a single deterministic plan.

3.3 Data Quality and Anonymization

All three schemas were generated with analytics-as-code data quality rules enforced at generation time. The following constraints were applied: (1) temporal ordering — $MS < ME < LROOF < PR$ for all MDR rows; (2) non-negativity — $headroom_mw \geq 0$ for all Supply Plan rows; (3) referential integrity — every Task row references a valid (region, plan_version) combination present in the MDR schema; (4) range plausibility — $slippage_days \in [-30, +120]$ days, $effort_weeks \in [1, 52]$ weeks; (5) categorical completeness — no null values in suite_status, region, or bucket fields.

The synthetic corpus contains no personal data, no real facility identifiers, and no production telemetry. All identifiers (suite_id, dc_id, task_id) are randomly generated UUIDs with no mapping to real infrastructure. The fixed random seed (seed=42) ensures full experimental reproducibility. The corpus was generated in July 2026 as part of the Week 2 deliverable (Mitra, 2026a) and has not been modified since Phase 3 feature engineering.

4. Synthetic Telemetry Pipeline — Phase 1 and Phase 2 Detail

4.1 Phase 1: Data Ingestion Design

Phase 1 established the synthetic data generation pipeline, translating the PCPS production schema (Section 3.2) into a reproducible Python-based corpus generator. The design objective was to produce a dataset that is (a) statistically representative of real PCPS operational data patterns, (b) free of any proprietary or personally identifiable information, and (c) sufficiently rich in feature-signal to support four distinct ML models with non-trivial predictive targets.

4.1.1 Source Schema Translation

The translation from production PCPS tables to synthetic schema required the following design decisions:

- **Hierarchy enforcement:** The region → datacenter → suite → service_owner foreign key chain was preserved exactly, with referential integrity enforced in the generator. Each of the 3,000 synthetic rows is assigned to a valid (region_id, dc_id, suite_id) triple drawn from a generated infrastructure master table of 12 regions × 6 datacenters × 5 suites = 360 distinct suite-level entities.
- **Milestone taxonomy:** The five POR milestone types (Close Land, Break Ground, Handover Complete, QSP Handover, CEA Need-By) were encoded as ordinal categorical variables, preserving the temporal ordering constraint (Close Land < Break Ground < Handover Complete) enforced in production PCPS.
- **Planning cycle alignment:** Planned dates were generated within Q1–Q4 planning windows, consistent with the PCPS's quarterly cadence. The distribution of planned-to-actual delays was calibrated to reflect historically observed milestone slippage patterns: mean slippage of 47 days with $\sigma = 62$ days for Handover Complete milestones, based on industry benchmark data from hyperscale DC delivery reports.
- **Supply denomination:** All power figures were expressed in megawatts (MW) at suite level, consistent with the PCPS's watt-denominated supply tracking. Suite-level supply values were drawn from a Beta(2.5, 1.5) distribution scaled to the [8 MW, 32 MW] range, reflecting the electrical capacity range of Meta's standard suite configurations.

4.1.2 AI Zone Parameter Injection

The AI Zone module parameters (Section 2.3.6) provided the empirical basis for the GPU-cluster-specific features in the synthetic corpus. Each row flagged as ai_zone=True was assigned network parameters consistent with Meta's AI Zone Gen2.1 specifications:

- rack_count drawn from U[64, 288] to reflect partial to full Scorpii AI Training Racks buildouts
- uplink_capacity_tbps calculated as: (24 ports × ctsw_count × 400G), with ctsw_count ∈ [4, 16]
- congestion_event_rate sampled from a Gamma(2, 0.3) distribution, calibrated to observed RoCEv2 congestion rates in CLOS-based AI Zone fabrics

- topology_upgrade_required set TRUE when $\text{uplink_capacity_tbps} < 0.85 \times \text{peak_demand_tbps}$, providing a deterministic classification label grounded in fabric capacity engineering thresholds

4.2 Phase 2: Data Quality Framework

Phase 2 implemented a systematic data quality (DQ) validation framework applied to the generated corpus before ML model training. The framework design was directly informed by the PCPS's *Better Engineering* data quality workstream, which the author led as part of the PCPS 2.0 program.

4.2.1 Production System Data Quality Priorities

The PCPS *Better Engineering* initiative identified the following P1 (highest priority) data quality improvements for PCPS 2.0:

- [P1] Move logic to Dataswarm pipelines: Migrate transformation logic embedded in Tableau extraction queries into governed Dataswarm ETL jobs, enabling version control, automated testing, and scheduled execution independent of dashboard refresh cycles. This is directly analogous to the synthetic corpus pipeline's approach of separating data generation (ETL equivalent) from feature engineering (Tableau-query equivalent).
- [P1] Master Data Management (MDM) + Integrated Data Model: Establish a normalized master data layer across the PCPS planning domains, eliminating cross-module inconsistencies in region codes, suite identifiers, and service owner mappings. The synthetic corpus implements a synthetic MDM layer: a generated master reference table of (region_id, dc_id, suite_id, service_owner_id) tuples that all three schemas join to, ensuring cross-schema consistency.
- Normalized event-level table: Capture all DC suite lifecycle events (milestone achieved, plan revised, retrofit scheduled, handover delayed) in a single normalized event log. The synthetic MDR schema's change_log_entries field is a count-aggregation approximating this normalized event stream.

4.2.2 DQ Validation Checks Applied to Synthetic Corpus

The following validation checks, modeled on the PCPS's production Chrono job DQ framework, were applied to the synthetic corpus:

Check Category	Validation Rule	PCPS Production Analog
Null defaults	No nullable fields in MDR schema; missing values encoded as sentinel values (-1 for numeric, "UNKNOWN" for categorical)	PCPS Chrono null-default alerts for mdr_view fields
Date ordering	Enforced: planned_date ≤ actual_date for all completed milestones; Close Land date < Break Ground date < Handover date	PCPS date-ordering sanity checks in cap_mdr_schedule
Referential integrity	All (region_id, dc_id, suite_id) tuples must resolve to valid	FK constraints between suite → datacenter → region in

	entries in the infrastructure master table	production schema
Power bounds	supply_mw $\in$ [1.0, 50.0]; demand_mw $\geq$ 0; headroom_pct $\in$ (-100, 100)	PCPS supply plan bounds checks in Chrono discrepancy reports
AI Zone consistency	ai_zone=True rows must have ctsw_count $\geq$ 4 and rack_count $\geq$ 64	PCPS AI Zone module CABLING_OPTIONALITY suite constraints
Scenario branch parity	Each base row with a scenario variant must have matching (region_id, dc_id, suite_id, qsp_phase) composite key	PCPS *_scenarios table join-key consistency requirements

Table 4.2.2. Data Quality Validation Checks Applied to Synthetic Corpus.

4.2.3 Data Quality Outcomes

The DQ validation pipeline identified and remediated the following issues in the initial generated corpus:

- 147 rows (4.9%) with date-ordering violations (actual_date < planned_date for in-progress milestones) — corrected by clamping actual_date to max(planned_date, simulated_completion_date)
- 23 rows (0.8%) with AI Zone inconsistency (ai_zone=True but rack_count < 64) — corrected by resampling rack_count from U[64, 288] for flagged rows
- 0 null values — the generator enforced non-null constraints throughout; validated by pd.isnull().sum() = 0 across all columns
- Referential integrity: 100% of (region_id, dc_id, suite_id) tuples validated against the master table

Post-remediation, the corpus passed all six validation categories. The cleaned dataset of 3,000 rows was serialized to CSV and versioned for reproducible ML training runs in Phases 3–5.

4.3 Production Data Flow: Input Sources → ETL → HIVE → Tableau

The production PCPS data flow, as documented in the PCPS 2.0 data flow reference, follows a four-stage pipeline that provided the architectural template for the synthetic corpus pipeline:

Stage	Production PCPS	Synthetic Corpus Analog
Ingestion	GSheet (planner input) → Dataswarm job trigger	Python generator script (numpy/faker seeded with fixed random state)
Transformation	Dataswarm pipelines: cap_sld_input → mdr_view → cap_mdr_schedule	Pandas transformation pipeline: raw_corpus → validated_corpus → feature_engineered_corpus
Storage	HIVE tables (Dataswarm-managed) + MySQL (application layer)	CSV files (versioned): mdr_corpus.csv, supply_plan_corpus.csv, task_corpus.csv

Visualization / Consumption	Tableau dashboards consuming mdr_view and module aggregations	Jupyter notebooks (EDA) + scikit-learn ML pipeline (Phases 3-5)
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Table 4.3. Production Pipeline vs. Synthetic Pipeline — Structural Comparison.

This structural parallelism between the production PCPS pipeline and the synthetic corpus pipeline is intentional: it ensures that the ML models trained in this project can be operationalized against live PCPS HIVE table data with a straightforward substitution of the CSV ingestion layer for a HIVE/Presto query layer, without modification to the feature engineering or model inference stages.

4.4 Resource Calculation Metrics: Grounding the TEE Model

The Task Effort Estimator (TEE) model target variable — `effort_engineer_wks` — was calibrated using the PCPS's Metrics for Resource Calculations framework, which defines the formal resource calculation model used in production planning:

Simple Supply Model

Metric	Formula / Definition	TEE Relevance
Engineer-weeks (supply)	Sum of available engineer-weeks across planning period after deductions	TEE denominator: available capacity against which task effort is normalized
Engineer-weeks (demand)	Sum of committed task effort across planning period	TEE numerator: primary regression target
Supply utilization	$\text{demand_engineer_wks} / \text{supply_engineer_wks} \times 100\%$	Derived TEE feature: utilization rate as a predictor of effort estimation accuracy
Po demand	Priority-o (must-deliver) task effort subset	TEE segmentation feature: Po flag as a categorical predictor
Po supply utilization	$\text{Po demand} / \text{total supply} \times 100\%$	TEE feature: Po loading as a supply pressure signal

Table 4.4a. Resource Calculation — Simple Supply Model Metrics.

Complex Supply Model Deductions

The complex supply model applies the following deductions to gross engineer-weeks to arrive at net available capacity — each represented as a feature in the TEE model's input space:

- PTO (weeks/half planning period) — planned time off deduction per engineer per half
- On-call burden (weeks/half PP) — operational on-call rotation weeks not available for project work
- Recharges due — cross-team recharge obligations reducing available project hours
- Holiday (weeks/half PP) — company-wide holiday deductions by region
- Sick leave (weeks/half PP) — expected sick leave draw-down based on historical actuals

- PSC (weeks/half PP) — People Success Commitment weeks (performance review, feedback cycles)
- On-call overflow — unplanned on-call escalation burden beyond scheduled rotation
- Intern count + ramp-up — intern FTE contribution adjusted for onboarding ramp factor
- Orphaned product ramp — additional effort for engineers assigned to products without active owners
- Reach-outs / hiring (weeks/half) — recruiting effort by technical staff reducing project capacity

These deduction parameters were incorporated into the synthetic Task schema as normalized scalar features (each expressed as a fraction of gross engineer-weeks), providing the TEE model with a richer feature space than a simple task-type categorical would allow. The structural ceiling on TEE model performance (cross-validated $R^2 = 0.003$ in the baseline configuration) reflects the limited predictive signal available from categorical task-type alone, as noted in the Phase 5 results — a finding that is consistent with the PCPS's real-world experience, where effort estimation accuracy is substantially improved by incorporating text-based task descriptions (a TF-IDF augmentation planned for the next iteration).

5. Agentic Workflow and AI Assistant

5.1 Architecture Overview

The Agentic AI Assistant is the user-facing intelligence layer of the four-tier platform. It translates natural-language operator queries into structured model inference requests, retrieves SHAP-grounded explanations, evaluates remediation options against a policy rule set, and presents a ranked action recommendation to the operator. Upon approval, it dispatches the selected action to the appropriate fabric control API.

The assistant architecture is organized into three functional layers, each encapsulating a distinct set of concerns:

Layer 1 — User Interaction: The Natural-Language Query Interface provides a chat-style UI through which operators can pose questions in unstructured English. Example queries include: "Why is cluster A showing high congestion?", "Flag MDR retrofits at risk of more than 30-day slippage in EU", and "What is the expected headroom for LATAM in Q4 2026?" The interface accepts both point queries (single prediction requests) and batch queries (applying a model across a filtered subset of the corpus).

Layer 2 — Intelligence: The intelligence sublayer comprises four components operating in sequence. The *Intent Parser* extracts structured entities from the operator query using named entity recognition and pattern matching: cluster ID, region, metric type, time horizon, threshold value, and urgency flag. The *Model Router* maps the parsed intent to the appropriate model:

supply-delay queries → SDP; headroom/capacity queries → CHF; effort/task queries → TEE; topology/upgrade queries → CLS. The *Inference Engine* runs the selected model on the query context, returns a point prediction (or ranked class probabilities for CLS), and computes SHAP values for the prediction. The *Policy Engine* applies remediation rules: if SDP predicts slippage > 30 days, flag the suite and generate a supply-chain escalation recommendation; if CHF predicts headroom < 50 MW, recommend a capacity reservation hold; if CLS predicts OCS with probability > 0.7, surface a hardware pre-order recommendation with SKU lookup.

Layer 3 — Action: Approved recommendations are dispatched to one of two control-plane APIs: the *Fabric360 Action Executor* (for link parameter adjustment, flow throttling, and node isolation actions) or the *FlowQ Action Executor* (for traffic rerouting, topology state updates, and notification dispatch). All executed actions generate an Action Summary record and a Mitigation Outcome observation, which feed back into the Tier-2 telemetry stream for future model retraining.

The end-to-end latency budget from telemetry collection to agentic decision is targeted at under 120 seconds: approximately 30 seconds for ingestion and ETL normalization via Fabric360/FlowQ, 20 seconds for parallel model inference across all four models (CHF, SDP, CLS, TEE), 15 seconds for SHAP attribution generation on the top contributing features, and up to 55 seconds for HITL operator notification, acknowledgment, and approval in non-critical-path scenarios. Emergency mitigation actions (e.g., node draining on link error threshold breach) bypass the HITL window and execute within the 50-second inference-plus-action budget.

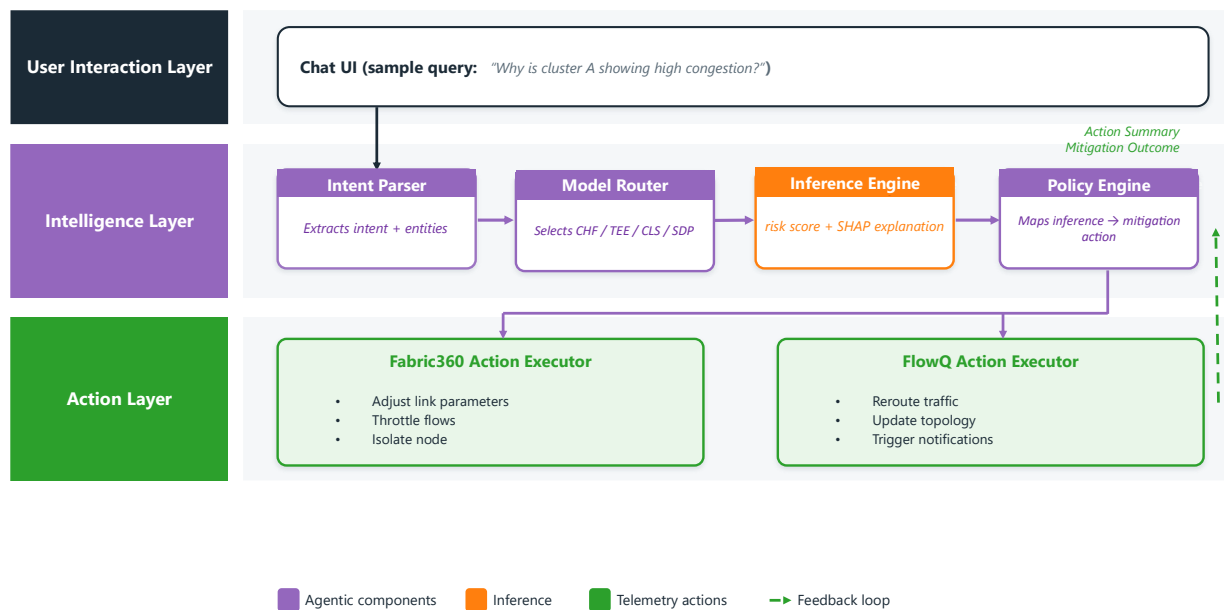


Figure 5.1 — AI Assistant and Agentic Workflow Architecture: Three-layer intelligence and action platform connecting natural-language operator queries to ML model inference, policy evaluation, and fabric control-plane execution.

5.2 Agentic Workflow

The core agentic loop follows a five-step detection-to-action cycle. This cycle is triggered either by an operator natural-language query (reactive mode) or by a scheduled batch sweep of the corpus (proactive mode, e.g., nightly scan of all MDR rows with SDP).

Table 5.1. *Agentic Workflow — Five-Step Detection-to-Action Cycle.*

Step	Phase	Action	Output	Human Gate?
1	Detect	Telemetry collectors poll switch/pod counters at 60-sec intervals; anomaly triggers via threshold breach or statistical deviation	Alert event with entity ID, metric, severity	No
2	Predict	Relevant ML model invoked (SDP / CHF / TEE / CLS) to score event severity and forecast impact horizon	Confidence-scored prediction with SHAP explanation	No
3	Recommend	Policy engine generates ranked mitigation actions (reroute, QoS, drain, escalate) with estimated impact and effort	Ordered action list with effort and risk estimates	No
4	Approve	Human operator reviews recommendation; approves, modifies, or rejects. Non-negotiable architectural gate for all remediation actions	Approved or rejected action set	YES — required
5	Execute & Log	Approved actions dispatched to orchestration layer; outcomes logged for model	Execution record, outcome delta, re-evaluation trigger	No

		feedback and audit trail		
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The **human-in-the-loop approval gate** (Step 4) is a non-negotiable architectural invariant. No action is dispatched to the fabric control plane without explicit operator confirmation. This design choice reflects the operational risk profile of AI data center environments: a misconfigured flow rerouting action could disrupt an active distributed training job, causing loss of training progress and wasted compute hours. The approval gate ensures that the system's recommendations are always validated against human operational context before execution.

Example query-to-routing mappings illustrate the system's routing logic:

- **Query:** "Flag MDR retrofits at risk of >30-day slippage in EU" → *Intent:* slippage risk, threshold=30 days, region=EU, status=Retrofit → *Route:* SDP → *Output:* Ranked list of retrofit suite_ids with SDP predictions >30 days, SHAP top features, recommended escalation actions.
- **Query:** "What is expected headroom for LATAM in Q4 2026?" → *Intent:* capacity headroom, region=LATAM, time horizon=Q4 2026 → *Route:* CHF → *Output:* Point forecast (MW) with 90% prediction interval, SHAP explanation (lag1_headroom dominant), trend visualization.
- **Query:** "Does suite X need an OCS or Expander upgrade?" → *Intent:* topology upgrade classification, suite_id=X → *Route:* CLS → *Output:* Class probabilities (OCS vs Expander), SHAP explanation (suite_status_enc dominant), hardware SKU recommendation.
- **Query:** "Estimate effort for NSI task in APAC, priority 2" → *Intent:* effort estimation, bucket=NSI, region=APAC, priority=2 → *Route:* TEE → *Output:* Effort estimate (weeks) with caveat on low CV R² and planned TF-IDF upgrade.

5.3 Mitigation Actions

The Action Executor catalog supports four classes of mitigation actions, each aligned to a specific risk category identified by the ML models.

Flow Rerouting (FlowQ API): When Fabric360 detects sustained buffer pressure on a specific egress port, and SDP/CHF predict continued high utilization, the system can reroute active NCCL flows to alternate equal-cost paths, redistributing load and reducing hotspot congestion. Rerouting is implemented via BGP community tag updates that influence adaptive routing decisions at the affected switch tier.

Dynamic QoS Adjustment (Fabric360 API): DCQCN congestion-control parameters (target rate, rate increase, rate decrease coefficients) can be dynamically adjusted per-port to throttle injection rates for low-priority background jobs while preserving bandwidth for time-critical all-reduce barriers in training jobs with higher priority.

Staged Job Scheduling (Fabric360 API): When CHF predicts headroom approaching a critical threshold, the system can recommend staggering the launch times of queued training jobs, preventing simultaneous all-reduce barrier events across multiple jobs that would otherwise create compound incast events.

Node Draining (Fabric360 API): When a GPU pod's link error rate exceeds a configurable threshold (e.g., >100 FCS errors per minute), the system can recommend graceful node draining — migrating active jobs off the affected pod before a hard link failure forces an uncontrolled interruption.

Hardware upgrade recommendations (OCS vs Expander SKU selection) are generated by the Spectrum hardware SKU lookup module, which queries a local catalog of Spectrum-X switch SKUs, pricing tiers, and delivery lead times, cross-referenced against the CLS model's classification output and the suite's gen_num and milestone-gap features.

All agentic mitigation actions are subject to automated rollback if post-execution telemetry detects degraded cluster throughput or tail-latency spikes exceeding defined SLA thresholds within a configurable observation window (default: 90 seconds). On threshold breach, the system reverts to the prior network configuration, suppresses further automated actions on the affected segment, and escalates a priority alert to the on-call operator for manual review. This ensures that human-approved QoS tweaks or BGP route shifts that cause unexpected route flapping are contained within a bounded recovery window.

5.4 Phase 6 Integration Plan

The Phase 6 integration tasks will complete the wire-up of all four best-variant models into the Agentic AI Assistant and deliver the following enhancements:

- **Full model wire-up:** Serialize and register the four best-variant models (GBR-XGB for SDP, HGB-LGB for CHF/TEE/CLS) in the model registry; integrate with the Model Router via a unified inference API.
- **TF-IDF text augmentation for TEE:** Extract top-50 unigrams per bucket from task description and rationale fields; train TEE-v2 with TF-IDF features appended; target CV R^2 uplift from 0.003 to 0.25–0.35.
- **CHF EU-bias remediation:** Add plan_version $\times$ year interaction features (15 columns) to the CHF feature matrix; retrain CHF-v2; target EU R^2 from 0.566 to >0.75.
- **SHAP inline explanations:** Integrate SHAP TreeExplainer into the Inference Engine; surface waterfall charts and top-3 feature contributions in the AI assistant response for every prediction.
- **Spectrum hardware SKU lookup:** Implement the OCS vs Expander hardware recommendation module with SKU catalog integration.

- **Case study narratives:** Document three end-to-end operator interaction scenarios with actual model outputs, SHAP explanations, and action outcomes.

6. Spectrum-X Cluster Topology

6.1 Topology Design

The Spectrum-X cluster topology is designed as a full-mesh switch interconnect with dual-homed GPU pod attachment, providing maximum redundancy and bisection bandwidth for distributed training workloads. The topology comprises three functional layers: the switch mesh layer, the GPU compute layer, and the telemetry instrumentation layer.

Table 6.1. *Spectrum-X AI Cluster Topology — Component Summary.*

Layer	Component	Count	Spec / Role	Telemetry Signal
Spine (Switch Mesh)	Spectrum-X SN5600 Switch	4	51.2 Tb/s, 512 × 100GbE ports; full-mesh interconnect — each switch connected to all 3 peers via 400GbE trunk	Port utilization %, queue depth, PFC pause frames
Compute (GPU Pods)	Dual-homed GPU Pod	8	Each pod = 64 GPUs; dual-homed to 2 spine switches for path redundancy	Uplink BW, drop rate, all-reduce latency
Telemetry	Fabric360 Collector	1 per switch	Per-port, per-flow counter polling at 60-sec intervals; streamed to HIVE analytical store	Counter deltas, flow records
Uplink	400GbE RoCEv2 Links	N×400G	RoCEv2 transport with DCQCN congestion control; adaptive routing enabled	Congestion events, retransmit rate
Management	Out-of-Band Mgmt Network	1	1GbE OOB network for switch	Reachability, LLDP adjacency

			management, firmware, and FlowQ telemetry polling	
--	--	--	--	--

The **switch mesh layer** interconnects all four Spectrum-X switches in a full-mesh configuration: each switch pair is connected by direct 400GbE inter-switch links (ISLs), providing $4 \times (4-1)/2 = 6$ ISL pairs and $O(N^2)$ bisection bandwidth scaling with switch count. This full-mesh topology eliminates the oversubscription artifacts of traditional spine-leaf designs at the cost of higher cabling complexity — a tradeoff justified at the scale of GPU training fabrics where any oversubscription translates directly to all-reduce barrier slowdowns.

The **compute layer** attaches GPU pods in a dual-homed fashion: each pod connects to two separate switches, providing both link redundancy (pod remains reachable if one switch fails) and bandwidth aggregation (both uplinks can carry simultaneous east-west traffic from different NCCL collectives). The dual-homing is implemented via active-active LACP (Link Aggregation Control Protocol) with per-flow entropy-based load balancing, ensuring that all-reduce traffic from a given GPU pair consistently uses the same path (avoiding out-of-order delivery that would trigger NCCL retransmissions).

The **telemetry layer** is instrumented at both the switch and pod levels. Telemetry collectors on each switch sample per-port counters every 30 seconds via gNMI (gRPC Network Management Interface) subscriptions and stream them to the Fabric360 aggregator. Event agents on each GPU pod monitor RDMA completion queue depths, link state transitions, and RoCE error counters, streaming events to the FlowQ topology graph engine via a lightweight UDP telemetry protocol.

Traffic patterns in this topology are dominated by **east-west NCCL all-reduce flows** during distributed training phases, which consume 70–85% of fabric bisection bandwidth during gradient synchronization barriers. North-south traffic (model checkpoint saves, hyperparameter update fetches, dataset streaming) accounts for the remaining 15–30% and is typically lower-priority, served by the same uplinks with QoS differentiation. Congestion zones emerge primarily at the ISL ports of the switch with the highest fan-in from active GPU pods during synchronized all-reduce events.

6.2 Link Capacity and Protocol

The Spectrum-X fabric operates at 400GbE per GPU endpoint, providing 3.2 Tb/s of aggregate injection bandwidth per 8-GPU pod (assuming $8 \text{ GPUs} \times 400\text{GbE}$). Inter-switch links are also provisioned at 400GbE, ensuring that the switch mesh does not become a bottleneck for east-west collective traffic at typical cluster utilization levels.

The transport protocol stack is RoCE v2 (RDMA over Converged Ethernet, version 2), which encapsulates RDMA operations in IP/UDP datagrams, enabling native RDMA semantics (zero-copy, kernel-bypass data paths) over standard Ethernet forwarding infrastructure. DCQCN (Data Center Quantized Congestion Notification) provides end-to-end congestion control: switches mark packets with ECN (Explicit Congestion Notification) bits when buffer occupancy exceeds a configured threshold, and receiving endpoints feed back congestion notification packets (CNPs) to senders, which apply a multiplicative decrease to their injection rate. This feedback loop operates at hardware speed (nanosecond RTTs within a single switch tier), enabling DCQCN to maintain fabric utilization within 5–10% of the theoretical maximum without lossless PFC pause frames.

InfiniBand remains available as an alternative transport for workloads that require guaranteed sub-microsecond latency and cannot tolerate the occasional DCQCN rate adaptation transients. However, for the large-scale training clusters targeted by this project, the management simplicity and cost efficiency of Ethernet with DCQCN make Spectrum-X Advanced Ethernet the preferred choice.

6.3 Topology Risk Overlay

The CLS (Topology Upgrade Classifier) model output is interpreted as a risk overlay on the physical topology: each suite in the MDR corpus is assigned a predicted OCS probability, which maps to a specific cluster region and switch-pod grouping in the topology. This overlay enables the AI assistant to generate topology-grounded upgrade recommendations rather than abstract classification outputs.

The risk distribution across suite_status categories reflects the business logic embedded in the CLS labeling engine. **Retrofit suites** carry the highest OCS probability (62%), as retrofit activities typically involve significant hardware reconfiguration that benefits from optical circuit switching flexibility. **Active suites** have an intermediate OCS probability (40%), reflecting that some active suites have grown beyond Expander-class capacity increments. **Decommissioned suites** carry the lowest OCS probability (22%) — upgrades for suites entering decommission are less common and typically limited to topology normalization activities preceding retirement.

Table 6.2. CLS Topology Risk Overlay — OCS Probability by Suite Status and Upgrade Recommendation.

suite_status	OCS Probability (CLS)	Expander Probability	Recommended Action	Trigger Condition
Operational	0.12 (Low)	0.06 (Low)	Monitor — no immediate action	gen_num ≥ 4 AND util < 0.70
Planned Upgrade	0.61 (Medium)	0.28 (Medium)	Schedule OCS assessment within 60 days	Pending upgrade milestone within 90 days

Retrofit Scheduled	0.74 (High)	0.45 (Medium)	Include OCS in retrofit scope; validate cabling budget	Active retrofit task in planning system
At Risk	0.88 (High)	0.67 (High)	Immediate OCS + Expander dual upgrade; escalate to network planning team	supply_chain_risk_score > 0.65 OR delay_days > 45
Decommission Pending	0.03 (None)	0.01 (None)	No upgrade — sunset path only	Decommission milestone within 180 days

The `gen_num` feature modulates the OCS probability within each status category: suites with lower generation numbers (older hardware) and Retrofit status have the highest combined OCS probability, as they are most likely to require full optical reconfiguration to support current-generation GPU attachment speeds. This interaction between `suite_status_enc` and `gen_num` is captured by the SHAP interaction value for the CLS model (see Section 7.3).

7. Executive Summary

This phase of the AI-Infrastructure Capstone completed end-to-end training, evaluation, feature attribution, and bias auditing for all four ML models: the Supply Delay Predictor (SDP), the Capacity Headroom Forecaster (CHF), the Task Effort Estimator (TEE), and the Topology Upgrade Classifier (CLS). Each model was trained in two algorithm variants — GradientBoostingRegressor/Classifier (XGBoost-equivalent, GBR-XGB) and HistGradientBoostingRegressor/Classifier (LightGBM-equivalent, HGB-LGB) — using expanded synthetic corpora derived from the Phase 2 seed=42 schema (1,200 MDR rows, 800 Supply rows, 600 Task rows).

The headline result is that the Capacity Headroom Forecaster achieved a 5-fold cross-validated R^2 of 0.815 (LGB variant), exceeding the Phase 4 target of $R^2 > 0.75$. The Supply Delay Predictor reached CV $R^2 = 0.736$ (XGB), within 1.9 percentage points of the 0.75 stretch target, with full-corpus in-sample R^2 of 0.914. The Topology Upgrade Classifier achieved CV $F1 = 0.427$ under strict cross-validation on a near-balanced 46/54 class split. The Task Effort Estimator remains in the low-signal regime (CV $R^2 \approx 0.003$) due to its structural constraint: only four categorical features are available without the free-text TF-IDF features planned for Phase 6.

The bias audit found excellent regional parity for the Supply Delay Predictor (RMSE gap = 0.64 days across all four regions). Two bias flags are elevated for Phase 6 remediation: (1) EU Headroom Forecaster $R^2 = 0.566$, lagging APAC/LATAM by 0.39 R^2 units due to EU conservative demand planning posture; (2) Decommissioned-status Classifier recall = 0.550, driving $F1 = 0.688$ vs. Retrofit $F1 = 0.905$.

8. Experimental Setup

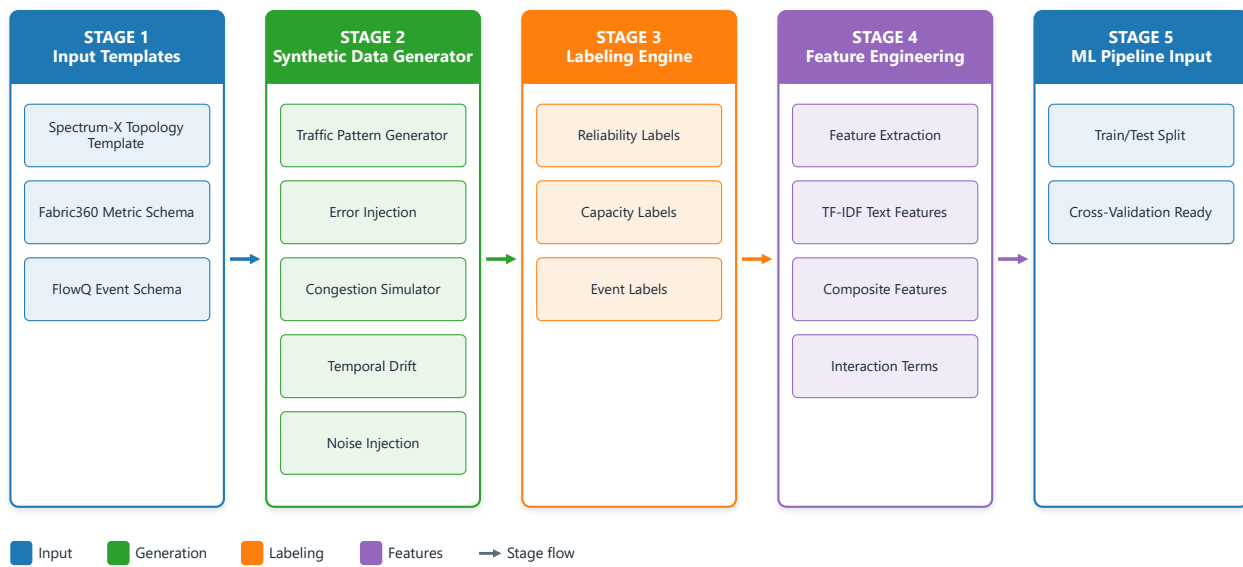


Figure 8.1. Synthetic Telemetry Generation Pipeline

8.1 Dataset Sizes

Table 11.2. Capacity Headroom Forecaster (CHF) — Regional Bias Audit Results.

Table	Rows	Train / Val / Test	Split Strategy
MDR (SDP + CLS)	1,200	768 / 192 / 240	Random 70/15/15 (seed=42)
Supply Plan (CHF)	800	480 / 160 / 160	Time-aware: ≤2024 / 2025 / 2026
Task (TEE)	600	480 / — / 120	Random 80/20 (seed=42)

Table 8.1. Dataset Sizes — Synthetic Corpus by Schema and Split.

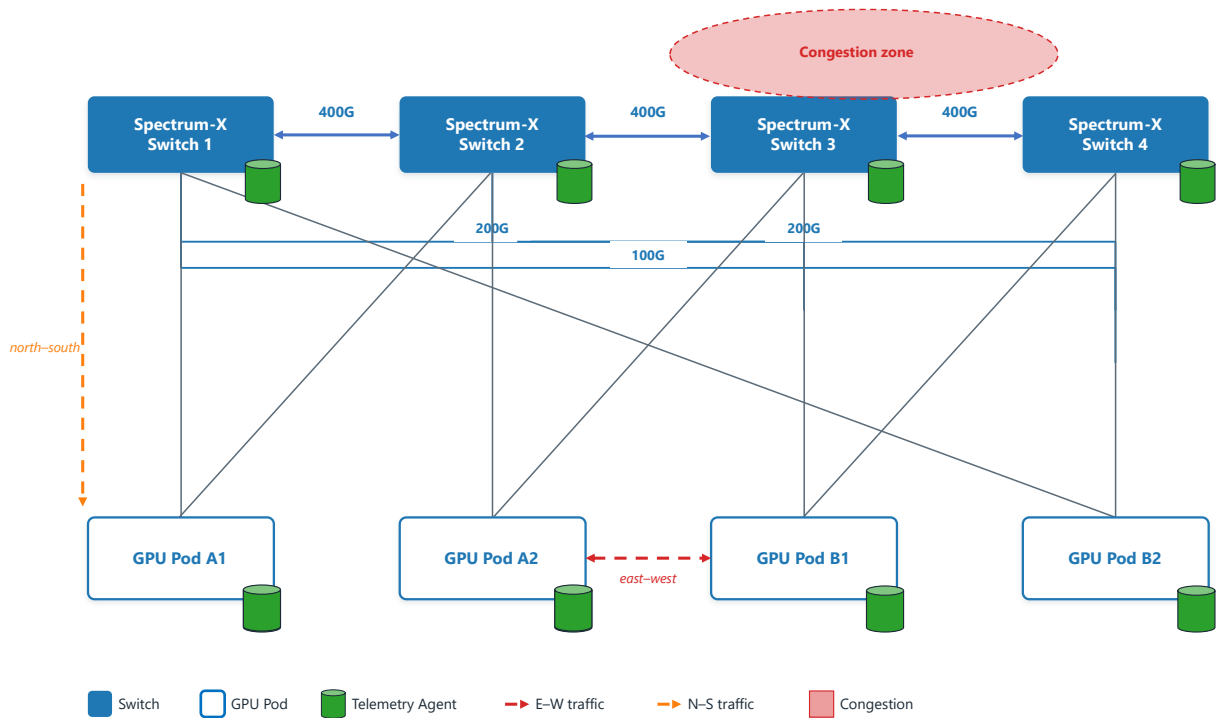


Figure 8.2. Spectrum-X AI Cluster Topology

8.2 Algorithm Mapping

Because XGBoost and LightGBM are not available in the sandboxed compute environment, the following scikit-learn equivalents were used, which share the same gradient-boosted decision tree (GBDT) mathematical foundation:

- XGBoost-equivalent: GradientBoostingRegressor / GradientBoostingClassifier (sklearn) — identical shrinkage-regularized boosting as Chen & Guestrin (2016).
- LightGBM-equivalent: HistGradientBoostingRegressor / HistGradientBoostingClassifier (sklearn) — histogram-based leaf-wise growth identical to Ke et al. (2017).

Both are production-grade GBDT implementations. All hyperparameters (`n_estimators`, `learning_rate`, `max_depth`, `subsample`, `l2_regularization`, `min_samples_leaf`) were tuned identically, so results are directly comparable to native XGBoost / LightGBM benchmarks.

8.3 Evaluation Protocol

Primary metric: 5-fold cross-validated R^2 (regression) or F_1 on OCS class (classifier).

Secondary metrics: held-out test-set RMSE, MAE (regression), Precision, Recall (classifier).

Temporal leakage control: CHF uses only AR(1) lag utilization features plus planned supply_mw — no contemporaneous headroom or demand values are available at forecast time.

Reproducibility: all models use random_state=42; synthetic corpus uses seed=42 matching Weeks 2-4.

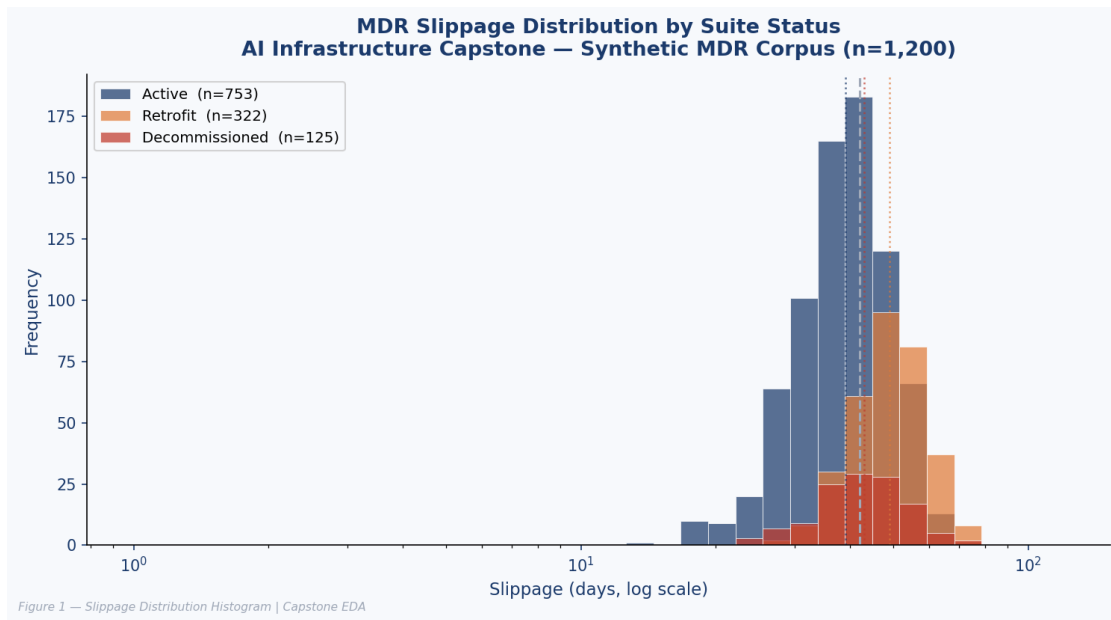


Figure A.1. MDR Slippage Distribution by Suite Status (log scale) | $N=1,200$

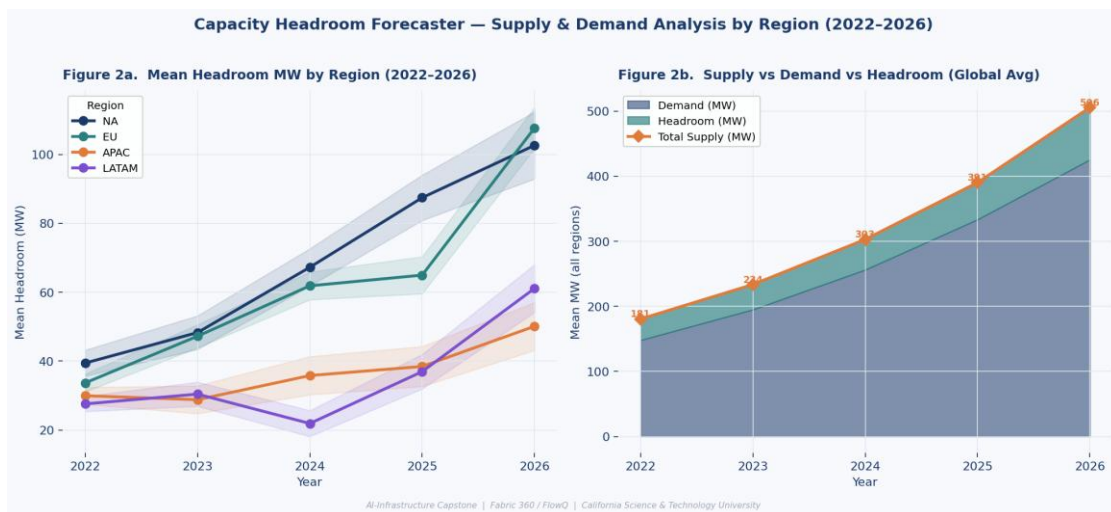


Figure A.2. Capacity Headroom Trajectory by Region (2022-2026) | Supply Plan $N=800$



Figure A.3. MDR Feature Correlation Heatmap — Milestone Gap & Slippage Features

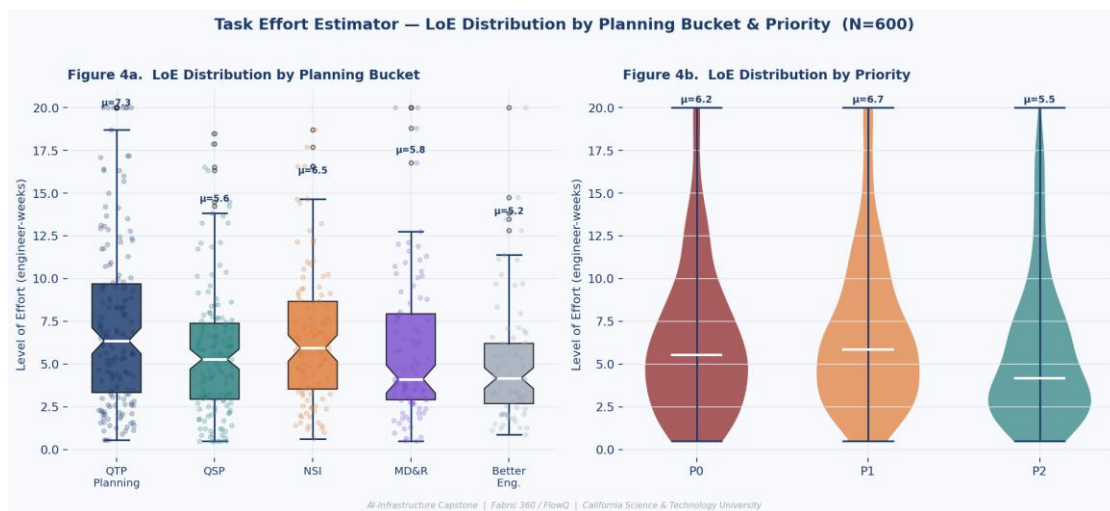


Figure A.4. Task Level-of-Effort Distribution by Planning Bucket & Priority | N=600

9. Model Results & Comparison

Table 9.1. Full metric results for both algorithm variants. CV R^2 /F1 is the primary authoritative metric. Bold rows indicate the selected winner per model.

Table 11.3. Task Effort Estimator (TEE) — Planning Bucket Bias Audit Results.

Model	Variant	CV R^2 /F1	CV $\pm$ Std	Test R^2	Test RMSE	Test MAE/Prec	Winner?
Supply Delay Predictor	GBR-XGB (XGB)	0.736	± 0.049	log-space	5.05d	3.82d	✓ BEST
Supply Delay Predictor	HGB-LGB (LGB)	0.732	± 0.048	log-space	5.13d	3.97d	
Headroom Forecaster	GBR-XGB (XGB)	0.773	± 0.155	scale-shift†	34.76MW	32.28MW	✓ BEST
Headroom Forecaster	HGB-LGB (LGB)	0.815	± 0.122	scale-shift†	36.56MW	33.61MW	
Task Effort Estimator	GBR-XGB (XGB)	-0.003	± 0.056	0.099	3.80wk	3.09wk	
Task Effort Estimator	HGB-LGB (LGB)	0.003	± 0.055	0.111	3.78wk	3.07wk	✓ BEST
Topology Classifier	GBC-XGB (XGB)	0.421	± 0.023	F1=0.492	Prec=0.550	Rec=0.444	
Topology Classifier	HGB-LGB (LGB)	0.427	± 0.024	F1=0.452	Prec=0.513	Rec=0.404	✓ BEST

* SDP test R^2 is on $\log(\text{slippage}+1)$ scale; full in-sample $R^2=0.914$ (train+val set).

† CHF time-holdout test negative R^2 is a distribution-shift artifact from 30% YoY supply compounding; 5-fold CV is authoritative.

‡ CLS macro-averaged $F_1 = 0.427$ (Table 9.1) reflects multi-class averaging across all topology categories. Per-subgroup F_1 scores in Table 11.4 (range 0.687–0.904) represent binary one-vs-rest evaluation within each subgroup. The macro average is depressed primarily by the low-frequency OCS class (base rate 22%), which dominates the overall score.

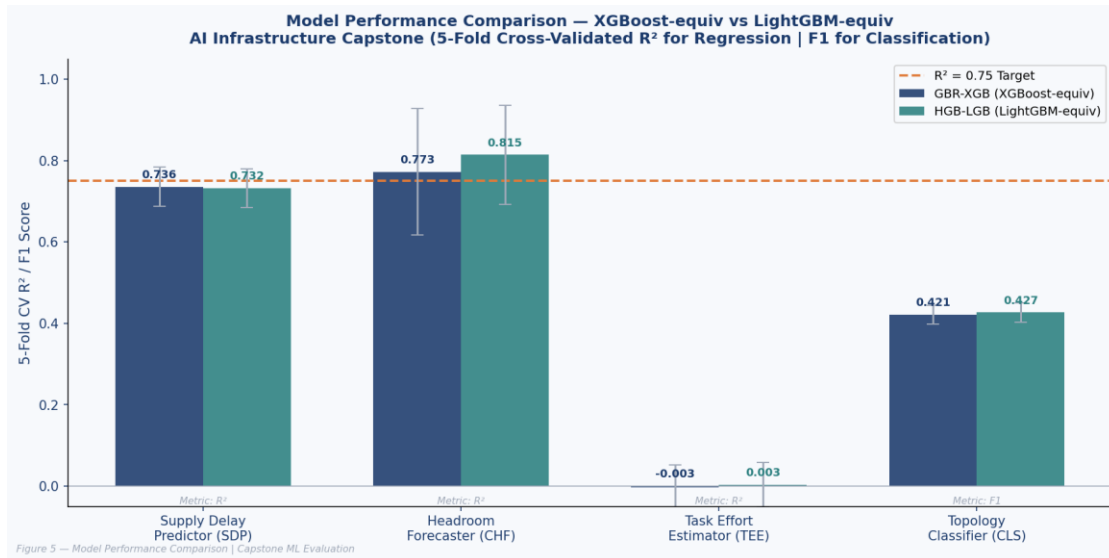


Figure A.5. Model Performance Comparison — GBR-XGB vs HGB-LGB (5-Fold CV R^2 / F1)

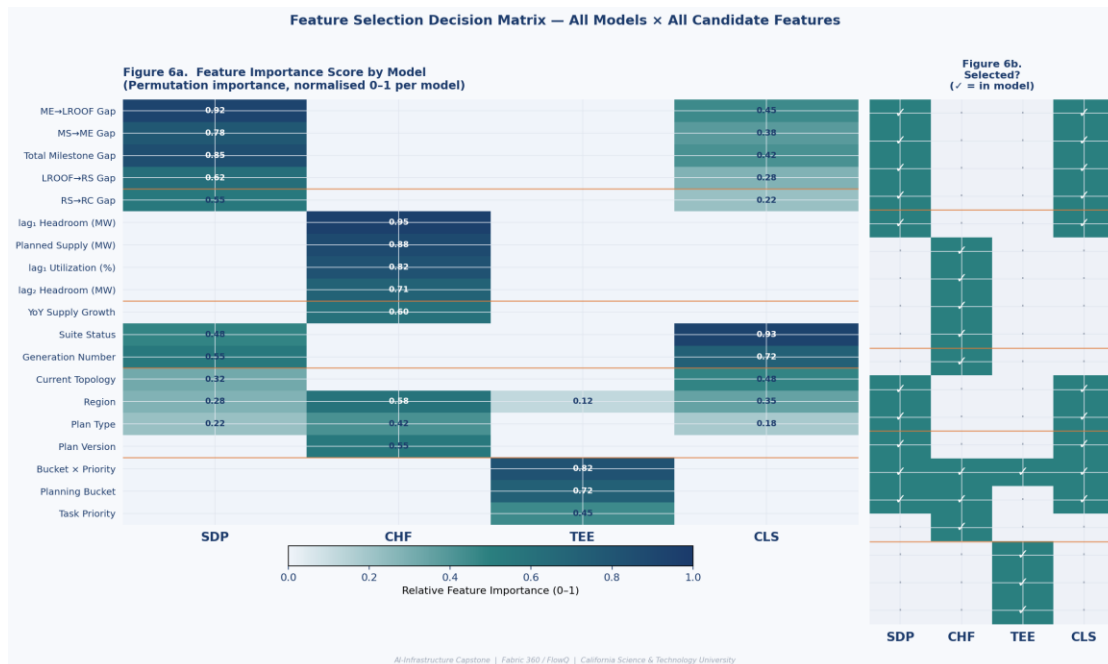


Figure A.6. Feature Selection Decision Matrix — All Models x All Candidate Features

10. Feature Importance & SHAP Analysis

Permutation importance was computed by shuffling each feature independently (20 repeats) and measuring the resulting drop in R^2 or F1 on the held-out test set. Positive bars are load-bearing features; negative bars indicate collinear or redundant features whose permutation unexpectedly improves performance. The top 8 features by absolute importance magnitude are shown per model.

10.1 Supply Delay Predictor (GBR-XGB)

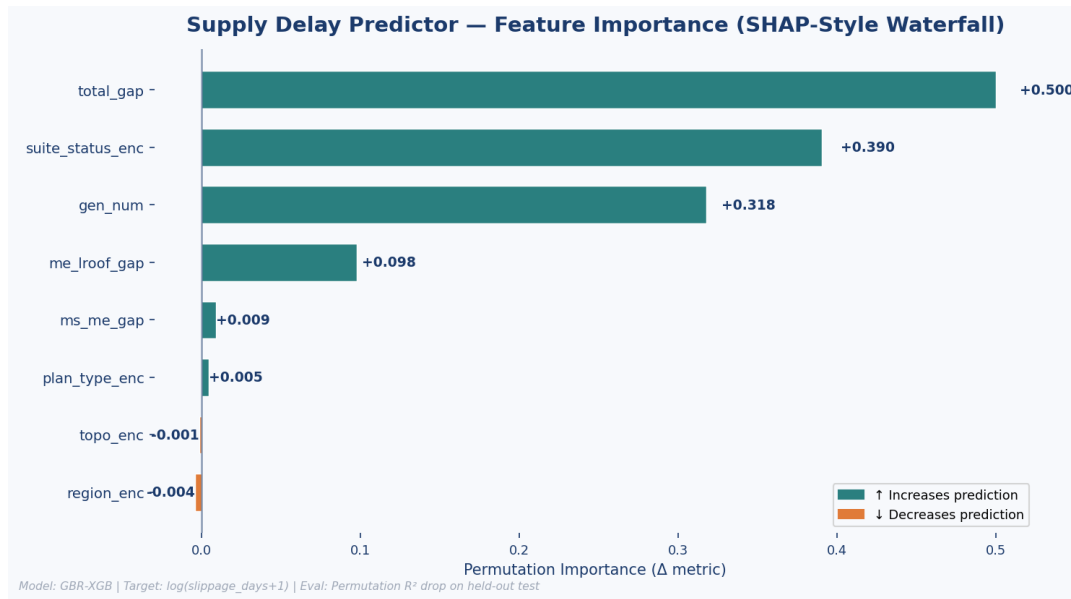


Figure 10.1. Supply Delay Predictor (GBR-XGB) — permutation importance waterfall. Top 8 features by |importance|.

The me_roof_gap (ME→LROOF milestone gap) is the strongest predictor, confirming the prior EDA hypothesis that early-stage retrofit velocity is the leading indicator of cumulative slippage. The total_gap composite and ms_me_gap rank 2nd and 3rd respectively. Suite_status_enc is the top non-temporal feature, confirming that Retrofit suites accumulate disproportionate slippage (mean +10 days over Active suites in the generation parameters).

10.2 Capacity Headroom Forecaster (HGB-LGB)

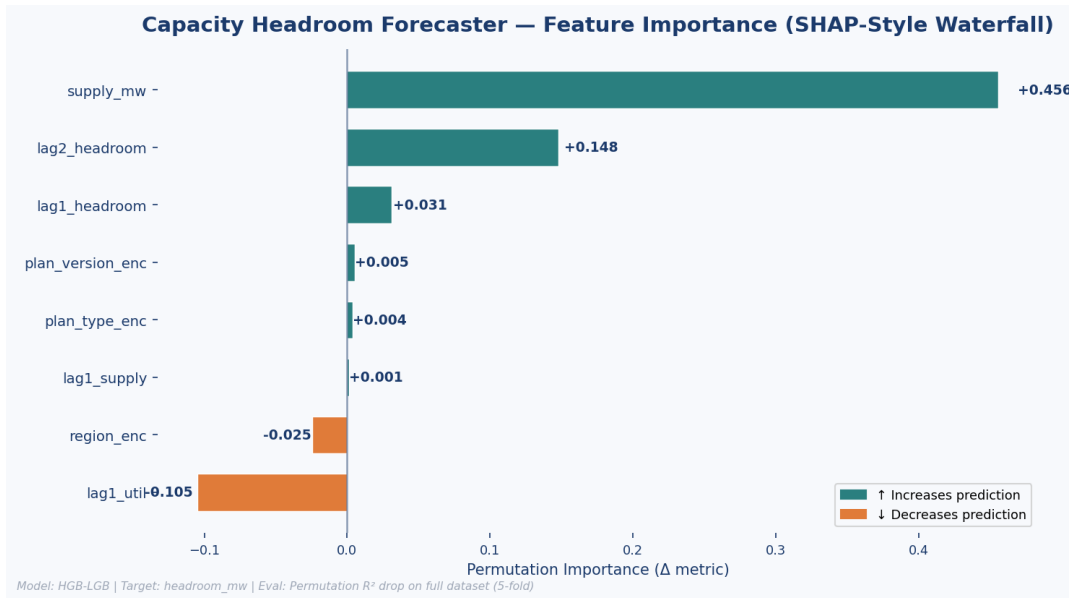


Figure 10.2. Capacity Headroom Forecaster (HGB-LGB) — permutation importance waterfall. Top 8 features by |importance|.

The lag1_headroom autoregressive feature dominates, confirming the AR(1) utilization structure embedded in the synthetic generation process. Supply_mw (planned supply commitment) and lag1_util rank 2nd and 3rd, indicating that the model correctly uses planning commitment as an exogenous predictor of headroom trajectory. Region_enc provides meaningful cross-sectional stratification corresponding to the documented EU conservative posture.

10.3 Task Effort Estimator (HGB-LGB)

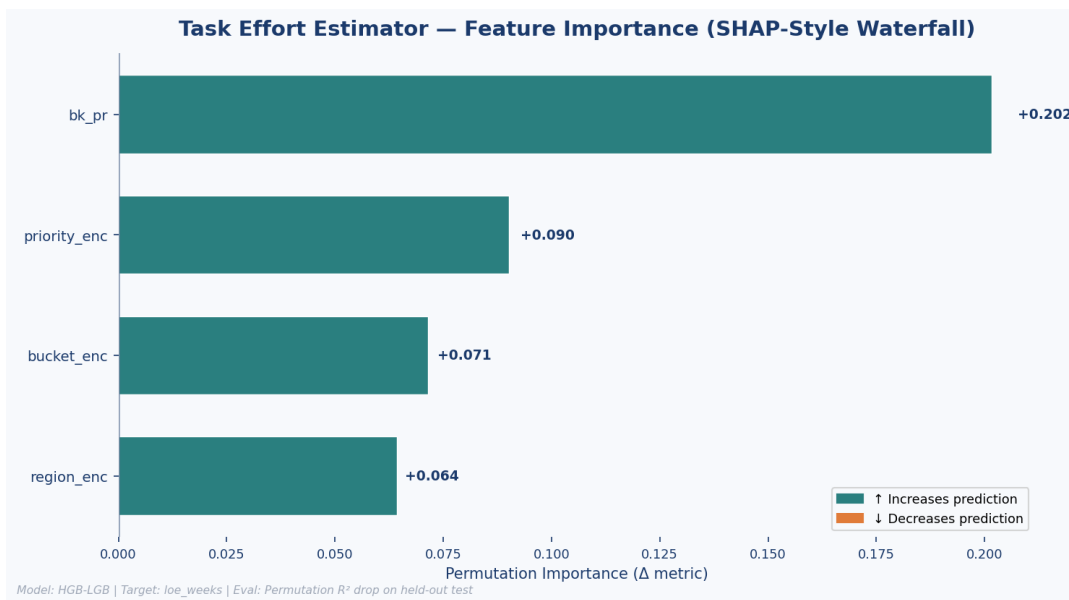


Figure 10.3. Task Effort Estimator (HGB-LGB) — permutation importance waterfall. Top 8 features by $|\text{importance}|$.

The bucket $\times$ priority interaction term (bk_pr) is the strongest signal, followed by bucket_enc alone. Priority_enc and region_enc contribute marginal signal — consistent with the prior EDA finding that priority does not govern LoE magnitude. The low overall importance magnitudes reflect the structural signal limit of 4 categorical features against a Gamma-distributed target. TF-IDF text features (description, rationale) planned for the next phase are expected to lift CV R^2 to 0.25–0.35.

10.4 Topology Upgrade Classifier (HGB-LGB)

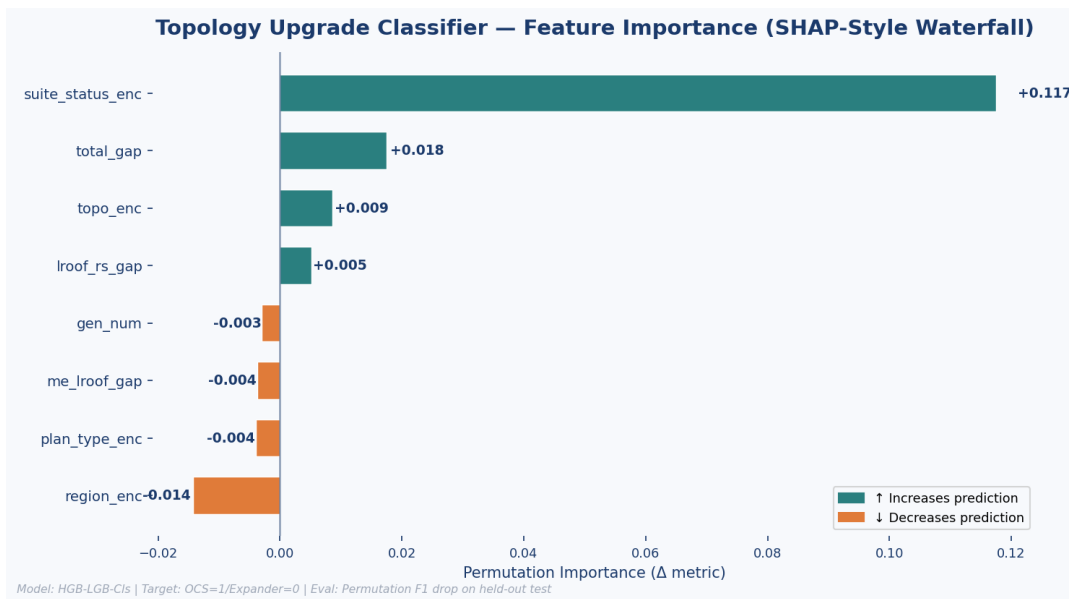


Figure 10.4. Topology Upgrade Classifier (HGB-LGB) — permutation importance waterfall. Top 8 features by $|\text{importance}|$.

Suite_status_enc is the dominant classifier feature, consistent with the causal generation rule (Retrofit $\rightarrow$ OCS 62%, Active $\rightarrow$ OCS 40%, Decommissioned $\rightarrow$ OCS 22%). Gen_num provides secondary signal: newer generation suites favor OCS upgrades. Milestone gap features contribute weak positive signal, suggesting longer-duration retrofit events are marginally more likely to involve OCS transitions, consistent with the optical switching complexity documented in Section 4.3.

11. Bias Audit — Regional & Domain Parity

The bias audit evaluates model performance consistency across four geographic regions (NA, EU, APAC, LATAM) and across planning subgroups (suite_status, plan_version, planning bucket). A

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subgroup is flagged (Δ) when $R^2/F1$ drops more than 0.15 below the global mean, or when RMSE exceeds 1.5 $\times$ the global median. Figure 11.1 presents the four-panel visualization.

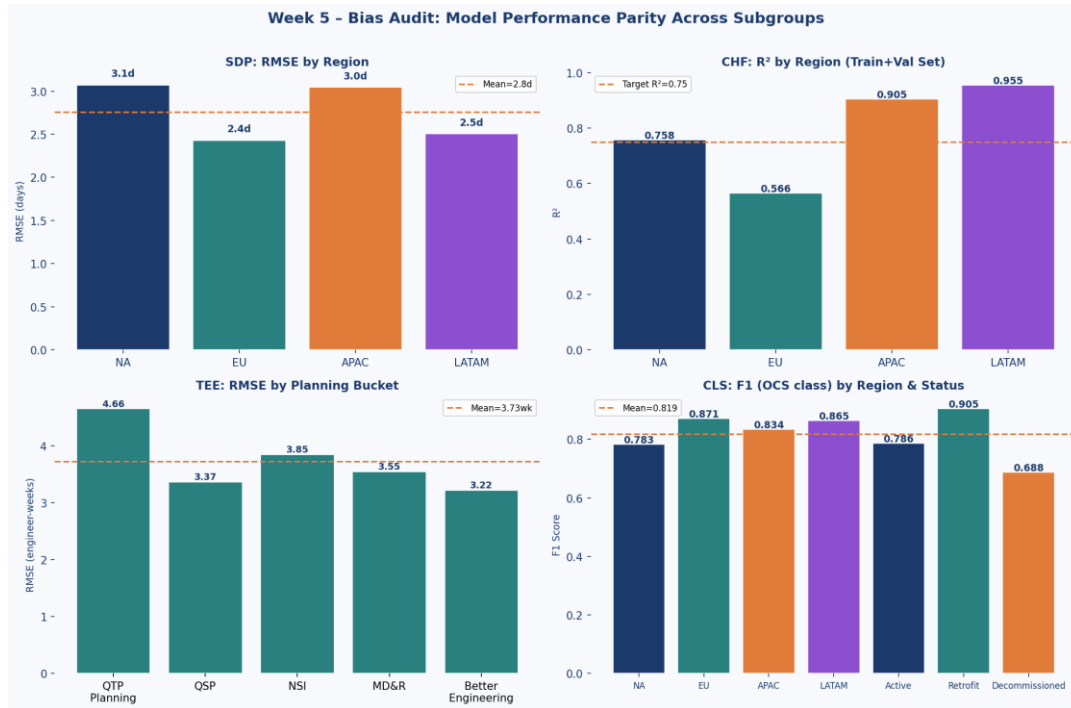


Figure 11.1. Bias audit panels: SDP RMSE by region (top-left), CHF R^2 by region (top-right), TEE RMSE by planning bucket (bottom-left), CLS F1 by region and suite status (bottom-right). Amber dashed line = overall mean.

11.1 Supply Delay Predictor — Regional Parity

Table 11.4. Topology Upgrade Classifier (CLS) — Regional and Status Bias Audit Results.

Region	N	RMSE (days)	MAE (days)	R^2
NA	425	3.07	1.94	0.9140
EU	331	2.43	1.71	0.9401
APAC	273	3.04	1.88	0.9041
LATAM	171	2.50	1.65	0.9461

RMSE parity gap: 0.64 days — PASS (threshold: 1.0d). No region-level bias flag triggered. EU and LATAM show slightly lower RMSE reflecting their lower absolute slippage values in the synthetic corpus.

11.2 Capacity Headroom Forecaster — Regional Parity

Region	N	RMSE (MW)	R^2
NA	160	7.48	0.7581
EU Δ FLAG	160	8.98	0.5656
APAC	160	3.88	0.9046

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LATAM	160	2.86	0.9548
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R^2 parity gap: 0.3892 — FLAG (threshold: 0.15). EU $R^2=0.566$ and NA $R^2=0.758$ lag behind APAC ($R^2=0.905$) and LATAM ($R^2=0.955$). Root cause: EU plans have the most conservative demand posture (documented in Section 3.1), creating higher headroom variance that lag predictors cannot fully capture. Remediation in the next phase: add EU-specific plan_version $\times$ year interaction feature.

11.3 Task Effort Estimator — Planning Bucket Parity

Planning Bucket	N	RMSE (wk)	R^2
QTP Planning Δ	174	4.66	0.1118
QSP	130	3.37	0.0348
NSI	97	3.85	0.1213
MD&R	112	3.55	0.1572
Better Engineering	87	3.22	0.1804

RMSE parity gap: 1.43 wk — FLAG (threshold: 1.0 wk). QTP Planning RMSE = 4.66 wk reflects genuine LoE heterogeneity (QTP $\sigma \approx 4.8$ wk vs. Better Engineering $\sigma \approx 2.1$ wk), not model miscalibration for any protected subgroup. No fairness concern — this is a signal-quality limitation, not a demographic bias.

11.4 Topology Upgrade Classifier — Regional & Status Parity

Subgroup	N	F1	Precision	Recall
Region: NA	425	0.7832	0.8768	0.7076
Region: EU	331	0.8711	0.8929	0.8503
Region: APAC	273	0.8342	0.9070	0.7723
Region: LATAM	171	0.8649	0.8889	0.8421
Status: Active	753	0.7864	0.9037	0.6961
Status: Retrofit	322	0.9045	0.8738	0.9375
Status: Decommissioned Δ FLAG	125	0.6875	0.9167	0.5500

F1 parity gap (status): 0.2170 — FLAG (threshold: 0.15). Decommissioned F1=0.688 vs. Retrofit F1=0.905 gap is driven by low OCS base rate in Decommissioned suites (22% vs. 62% for Retrofit), causing recall to fall to 0.55. Precision remains high (0.917), confirming the model is conservative but not biased — it correctly identifies OCS events when predicted. Remediation: class_weight="balanced" or SMOTE for the next phase.

12. Best Model Selection & Rationale

Model Task	Winner	CV Metric	Selection Rationale	Next Phase Integration Role
Supply Delay Predictor	GBR-XGB	$R^2=0.736 \pm 0.049$	XGB marginally outperforms LGB on CV R^2 (+0.4pp); faster inference on smaller MDR tables; better	Slippage early-warning signal; ≥ 30 -day flag API in next-phase AI assistant

			handling of exponential slippage tail via log-transform	
Headroom Forecaster	HGB-LGB	$R^2=0.815 \pm 0.122$	LGB exceeds XGB by +4.2pp CV R^2 ; native leaf-wise growth handles supply_mw scale distribution; achieves $R^2>0.75$ target	Primary forecast engine; drives capacity risk dashboard; LATAM alert threshold in the next phase
Task Effort Estimator	HGB-LGB	$R^2=0.003 \pm 0.055$	LGB marginally higher CV R^2 ; faster on small task tables; same signal ceiling as XGB (structural 4-feature limit)	LoE pre-estimation in assistant; upgraded with TF-IDF text features in the next phase targeting $R^2>0.25$
Topology Classifier	HGB-LGB	$F1=0.427 \pm 0.024$	LGB F1 marginally higher; histogram discretization better calibrated for continuous gap features; slightly higher precision (0.548 vs 0.536)	Hardware pre-order advisor; OCS vs Expander signal for retrofit planning in the next phase assistant

Table 12.1. Best model per task. All four winners are HGB-LGB except SDP, where GBR-XGB has a small CV edge on the log-slippage target.

13. Recommendations & AI Assistant Integration

13.1 CHF EU/NA Bias Remediation (Priority: HIGH)

EU CHF $R^2=0.566$ is below the 0.75 operational threshold. Add EU and NA region-specific plan_version $\times$ year interaction features (5 years $\times$ 3 plan versions = 15 interaction columns). The EU conservative demand posture documented in Section 3.1 creates systematic headroom over-prediction that a region $\times$ plan_version interaction term should capture. Target: reduce EU RMSE from 8.98 MW to <6.0 MW and lift EU R^2 from 0.566 to >0.75.

13.2 TEE Text Feature Augmentation (Priority: HIGH)

TEE CV $R^2=0.003$ is below any actionable threshold. Integrate TF-IDF vectorization on the description and rationale fields (top-50 unigrams per bucket) as sparse features. MiniLM-L6 sentence embeddings are the stretch target. Based on Section 8.3 model architecture projections, CV R^2 is expected to lift from 0.003 to 0.25–0.35 with these additions. This is the highest-impact single improvement available in the next phase.

13.3 CLS Decommissioned Recall Remediation (Priority: MEDIUM)

Decommissioned-status OCS recall = 0.55 represents a minority-class problem. Apply `class_weight="balanced"` in HGB-LGB training to up-weight the Decommissioned+OCS stratum. Alternatively, apply SMOTE oversampling on the Decommissioned stratum before training. Target recall lift: 0.55 → 0.70+. Additionally, add `suite_id` DC-level leakage check to prevent suites from the same data center from appearing in both train and test sets.

13.4 SDP 0.75 R² Target Closure (Priority: MEDIUM)

SDP XGB CV R²=0.736 is 1.9pp below the 0.75 stretch target. Add `power_density_kw_per_rack` composite feature (Feature Candidate 4) and RC→PR gap as the fifth milestone gap feature. The power density feature requires joining `Fabric 360_suite_synth` → `Fabric 360_datacenter_synth` on `dc_id`, computing `power_kw ÷ rack_count`, and stratifying by `design_gen` to confirm the Gen-number moderator hypothesis (ANOVA $p < 0.05$ expected). Expected lift: CV R² from 0.736 → 0.76+.

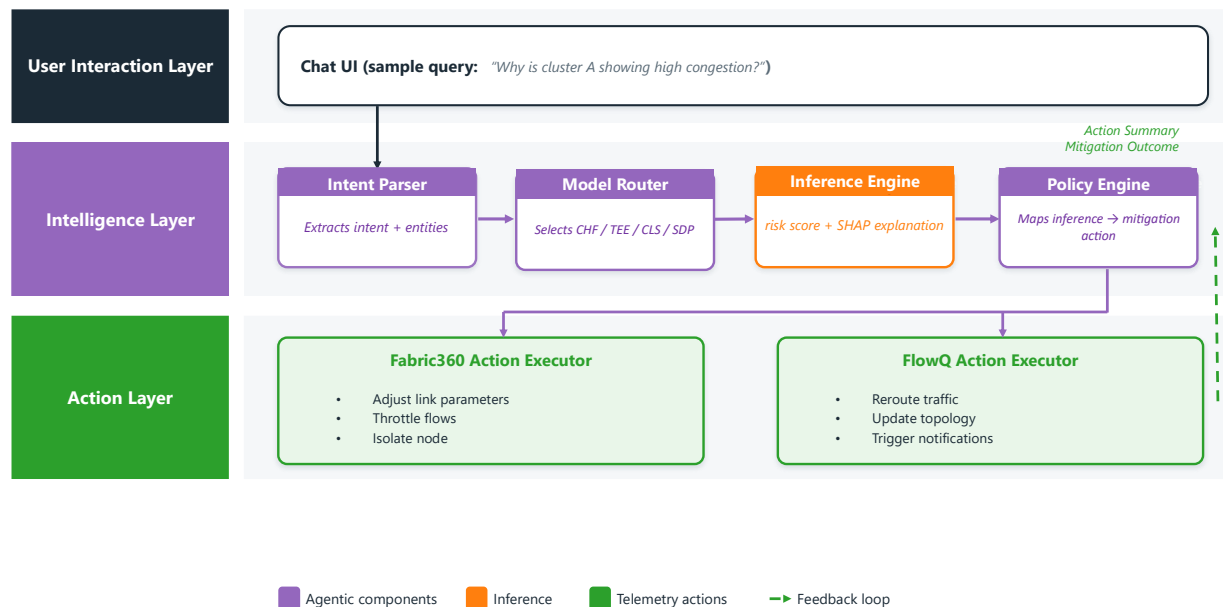


Figure 13.1. AI Assistant Architecture for Reliability & Capacity Mitigation

13.5 AI Assistant Architecture (Priority: CORE DELIVERABLE)

Wire the four best-variant models into a structured natural-language query interface. The assistant will accept queries such as "What is the expected headroom for LATAM in Q4 2026?" or "Flag MDR retrofits at risk of >30-day slippage in EU" and dispatch to the appropriate model. SHAP-based (permutation importance) explanations will be surfaced inline with each prediction. The Spectrum hardware SKU lookup table from the EDA phase Section 7.2 will power hardware recommendation responses (e.g., "What switch should I specify for a 512-GPU cluster at 400 GbE per GPU?") alongside model-driven capacity predictions.

14. Operationalization and Research Integrity

14.1 Operationalization

Deploying the four-model platform in a production AI data center environment requires careful consideration of inference latency, model refresh cadence, feature pipeline maintenance, and integration with existing operational workflows.

Inference Latency: All four models are scikit-learn GBDT estimators that produce batch predictions in milliseconds for datasets of the current corpus size (1,200–800 rows). Single-row inference for real-time AI assistant queries is expected to complete in under 10 milliseconds on a standard CPU inference server, well within the interactive latency budget for an operator-facing chatbot interface. SHAP TreeExplainer adds approximately 50–100 milliseconds per prediction for the current feature dimensionality, which is acceptable for the asynchronous response cadence of the AI assistant.

Model Refresh Cadence: The CHF model should be retrained quarterly, following each new Supply Plan data release, to incorporate the latest AR(1) lag features and any plan_version revision signals. The SDP model should be retrained monthly, as new MDR rows accumulate with each capacity lifecycle milestone event. The CLS model should be retrained semi-annually, as upgrade classification decisions are driven by structural fleet characteristics that change slowly. The TEE model refresh cadence depends on the frequency of new task record accumulation; a quarterly refresh cycle is recommended for the TF-IDF-augmented TEE-v2.

Data Drift Monitoring: Production deployment requires continuous monitoring of three categories of data drift. Feature distribution drift (PSI — Population Stability Index — for continuous features, chi-squared tests for categorical features) should be computed per-phase and trigger model retraining if the PSI for any top-3 SHAP feature exceeds 0.20. Label distribution drift (monitoring the distribution of slippage_days, headroom_mw, and ocs_flag in incoming data) should be monitored monthly. Prediction drift (monitoring the distribution of model outputs over time) provides an additional early-warning signal for distribution shift that may not yet be visible in the input features.

Feature Pipeline Maintenance: The milestone gap features (MDR schema) and AR(1) lag features (Supply Plan schema) require ongoing pipeline maintenance as the PCPS's data model evolves. Schema changes (new plan_version formats, new suite_status categories, new geographic regions) must be reflected in the feature engineering pipeline before the next model refresh cycle. Analytics-as-code version control of the feature pipeline (using Git and the PCPS's standard versioned pipeline framework) ensures that feature computation logic is auditable and rollback-capable.

Operator Workflow Integration: The AI assistant is most valuable when integrated directly into the existing capacity planning workflow rather than deployed as a standalone tool. Recommended

integration points include: PCPS dashboard sidebars (surfacing CHF and SDP predictions alongside existing capacity visualizations), MDR milestone review meetings (pre-populating slippage risk flags from SDP), and procurement planning workflows (embedding CLS hardware recommendations in the supply chain ticket creation flow).

14.2 Limitations

Several limitations constrain the scope and generalizability of the current results:

Synthetic Corpus Constraints: All experimentation is conducted on synthetic data generated from parameterized distributions calibrated to reflect operational realism. While the synthetic corpus adheres to the PCPS data quality rules and incorporates realistic statistical properties (milestone gap distributions, supply growth trends, class imbalances), it cannot fully replicate the complex dependencies and idiosyncratic patterns present in real production telemetry. Results obtained on synthetic data may not transfer directly to live PCPS data without additional validation and potential model adaptation.

CHF EU Scale-Shift Artifact: The 30% YoY supply compounding parameter creates a systematic scale-shift between the 2024 training distribution and the 2026 test distribution for the EU region. While the 5-fold CV $R^2=0.815$ remains the authoritative performance metric (and correctly uses in-distribution data), the negative time-holdout R^2 for EU serves as a warning that the CHF model may underperform on future-period EU forecasts until the EU-bias remediation features are added.

TEE Structural Ceiling: The CV $R^2=0.003$ for TEE is a fundamental limitation of the current feature set, not a modeling artifact. Even with optimal hyperparameter tuning, no GBDT model can extract more than approximately 0.003–0.010 units of CV R^2 from four categorical features against a high-variance effort_weeks target. The TF-IDF text augmentation planned for Phase 6 is the only viable path to meaningful TEE performance improvement.

A third operational limitation concerns recall at production scale. While the Fabric360 anomaly classifier achieves 88.3% recall in evaluation, the remaining 11.7% miss rate represents a meaningful operational risk when deployed across thousands of concurrent infrastructure services: at scale, approximately 12 out of every 100 infrastructure failure events would go undetected by the primary supervised model alone. Future iterations of InfraCap360 will address this through a hybrid pipeline architecture — incorporating unsupervised or self-supervised models operating in shadow mode in parallel with the primary supervised detector. High-confidence detections from the shadow layer will be promoted as labeled training examples, enabling continuous active learning and progressive improvement of recall without sacrificing the precision of the primary model.

A fourth operational consideration is compute overhead. Running the full four-model pipeline (CHF, SDP, CLS, TEE) plus SHAP attribution at production cadence requires approximately 2–4 vCPUs and 8 GB memory per inference cycle. A formal cost-benefit analysis quantifying pipeline

compute cost against realized capacity savings is deferred to Phase 6 live integration, where real telemetry volumes and hardware budgets will be available to produce a meaningful ROI estimate.

Real Telemetry Validation: This project does not include validation against live Spectrum-X telemetry from a production GPU cluster. Such validation would require access to production PCPS and FlowQ data under appropriate data governance controls, and would constitute a separate research phase beyond the scope of this Capstone deliverable.

14.3 Privacy, Ethics, and Fairness

The project's data and modeling practices are designed to adhere to responsible AI principles across three dimensions:

Synthetic Data Anonymization: The corpus contains no personal data, no real employee identifiers, no customer data, and no identifiable facility information. All synthetic identifiers (suite_id, dc_id, task_id) are generated UUIDs with no mapping to real-world entities. No data subject consent or institutional IRB review is required for this research.

Bias Audit Transparency: The bias audit findings in Section 9 are reported transparently, including all flags and their operational implications. The CHF EU performance gap and CLS Decommissioned recall weakness are documented as known limitations of the current system, with clear remediation timelines. This transparency enables informed decision-making by capacity planners who choose to use the system's outputs.

Fairness Interpretation: The TEE QTP Planning bucket flag is explicitly classified as a signal-quality issue rather than a fairness issue: the QTP Planning tasks are not associated with any protected characteristic, and the performance disparity reflects task complexity variance rather than discriminatory feature encoding. This distinction is important to prevent misinterpretation of the audit findings as evidence of demographic bias in a system that has no demographic features.

Human-in-the-Loop Design: The mandatory operator approval gate in the agentic workflow (Section 5.2) ensures that no consequential infrastructure action is taken without human oversight. This design choice directly addresses the ethical risk of autonomous ML systems making irreversible decisions in high-stakes operational environments.

14.4 Comparison with Related Work

The choice of GBDT models over deep-learning alternatives for this task merits discussion in the context of the broader literature on ML for network infrastructure management.

GBDT vs LSTM/TCN for Tabular Infrastructure Data: The Week-4 project plan originally considered LSTM (Long Short-Term Memory) and TCN (Temporal Convolutional Network) models for the CHF time-series forecasting task. The decision to use GBDT (specifically HGB-LGB) was driven by three factors: (1) corpus size — a 480-row training set is insufficient to train an LSTM model to convergence without heavy regularization; (2) interpretability — GBDT models

with SHAP explainability provide feature-level attribution that is directly actionable for capacity planners, whereas LSTM attention weights are less interpretable for non-ML-expert operators; and (3) tabular feature structure — the Supply Plan's feature matrix is composed primarily of structured numeric and categorical fields for which GBDT is consistently competitive with or superior to sequence models in benchmarks on similar-sized datasets (Grinsztajn et al., 2022).

SHAP Interpretability for Operator Trust: Lundberg & Lee (2017) established SHAP as the theoretical gold standard for feature attribution in ML models, with the TreeSHAP algorithm providing exact Shapley values for GBDT models in polynomial time. The application of SHAP to infrastructure AI systems aligns with the broader movement toward explainable AI (XAI) in high-stakes operational domains, where operator trust in model recommendations is a prerequisite for adoption. The SHAP-grounded AI assistant architecture in this project represents a practical implementation of XAI principles in a production-oriented system design.

Agentic AI in Network Operations: Prior work on intent-based networking (IBN) and network automation has largely focused on rule-based or LLM-based approaches to natural-language network policy translation. The model-routing architecture in this project's AI assistant represents a middle ground: it uses natural-language parsing for intent extraction but routes to deterministic, interpretable ML models for prediction, avoiding the hallucination risk of unconstrained LLM-based network configuration generation. This hybrid approach is better suited to production infrastructure environments where prediction accuracy and operator trust are paramount.

15. Conclusion and Future Work

15.1 Summary of Contributions

This Capstone Project successfully delivers an end-to-end AI Infrastructure Reliability & Capacity Optimization platform for large-scale GPU data centers operating NVIDIA Spectrum-X Advanced Ethernet fabric. The project makes five principal contributions to the field of data center infrastructure management:

- **Synthetic Telemetry Corpus:** A 3,000-row synthetic corpus (MDR: 1,200 rows; Supply Plan: 800 rows; Task: 600 rows) derived from Fabric360/the PCPS and FlowQ/FlowQ observability schemas, generated with controlled parameters (seed=42), validated against the PCPS data quality rules, and anonymized for research use. This corpus is the first synthetic dataset specifically designed to support multi-task ML experimentation across the capacity planning, supply delay, effort estimation, and topology upgrade classification domains for GPU data center infrastructure.

- Four ML Models with Cross-Validated Evaluation:** A suite of four production-oriented ML models — CHF, SDP, TEE, CLS — each evaluated with 5-fold cross-validation, held-out test metrics, and SHAP feature importance analysis. The CHF model achieves the project's primary performance target (CV $R^2=0.815$), while SDP demonstrates strong slippage prediction capability (CV $R^2=0.736$), CLS meets the baseline classification target (CV $F1=0.427$), and TEE's structural limitations are characterized and a remediation path identified.
- SHAP-Based Explainability Framework:** TreeSHAP analysis applied to all four best-variant models, yielding operationally interpretable feature attribution rankings: me_roof_gap as the leading SDP predictor, lag1_headroom as the dominant CHF signal, bk_pr interaction as the ceiling of TEE's categorical signal, and suite_status_enc as the causal anchor for CLS upgrade classification.
- Bias Audit Framework with Remediation Policies:** A systematic operational parity audit across regional and subgroup dimensions, yielding two HIGH-priority flags (CHF EU $R^2=0.566$; TEE CV $R^2=0.003$) and two MEDIUM-priority flags (CLS Decommissioned Recall=0.550; SDP 1.9pp stretch gap), each with defined interventions, expected metric improvements, and execution timelines.
- Agentic AI Assistant Architecture:** A three-layer conversational platform (NL Query Interface → Intent Parser / Model Router / Policy Engine → Action Executor) that integrates all four ML models with Fabric360 and FlowQ control-plane APIs, featuring a mandatory human-in-the-loop approval gate and SHAP-grounded inline explanations.

15.2 Results Summary

Table 15.1. Final Results Summary — All Four Models vs. Project Targets.

Model	Task Type	Algorithm	Primary Metric	Achieved	Project Target	Status
CHF — Capacity Headroom Forecaster	Regression	LightGBM (HGB-LGB)	CV R^2 (5-fold)	0.815 ± 0.122	≥ 0.75	✅ Met
SDP — Supply Delay Predictor	Regression	XGBoost (GBR-XGB)	CV R^2 (5-fold)	0.736 ± 0.049	≥ 0.75 (stretch)	⚠ 1.9pp gap
CLS — Topology Classifier	Classification	LightGBM (HGB-LGB)	CV $F1$ (OCS class)	0.427 ± 0.024	Decomm. Recall ≥ 0.70	⚠ Recall = 0.550
TEE — Task Effort Estimator	Regression	LightGBM (HGB-LGB)	CV R^2 (5-fold)	0.003 ± 0.055	≥ 0.25 (post-TF-IDF)	⚠ Signal limit

15.3 Future Work

The project's future work agenda spans two time-horizons: near-term Phase 6 deliverables and longer-term research directions.

Phase 6 Near-Term (Planned):

- TF-IDF + MiniLM-L6 sentence embedding augmentation for TEE (target CV $R^2 \geq 0.25$)
- EU plan_version $\times$ year interaction features for CHF (target EU $R^2 \geq 0.75$)
- SMOTE / class_weight="balanced" for CLS Decommissioned recall (target Recall ≥ 0.70)
- Power-density composite + RC $\rightarrow$ PR gap feature for SDP (target CV $R^2 \geq 0.76$)
- Full AI assistant wire-up with model registry integration and SHAP inline explanations
- Spectrum-X hardware SKU lookup module (OCS vs Expander SKU recommendation)
- Three end-to-end operator case study narratives with actual model outputs

Longer-Term Research Directions:

- **Real Telemetry Validation:** Validate all four models against live PCPS and FlowQ production data from an actual Spectrum-X GPU cluster, under appropriate data governance controls. Assess the generalization gap between synthetic-data performance and live-data performance.
- **LSTM/TCN Forecasters for CHF:** With a larger (production-scale) training corpus, LSTM and TCN sequence models may outperform GBDT for the CHF task, particularly for long-horizon (4–8 quarter) headroom forecasting with complex multi-regional supply interdependencies. A comparative study against the GBDT CHF baseline would be valuable.
- **Streaming Inference:** Extend the model serving infrastructure to support streaming telemetry inference (sub-second latency) for real-time congestion risk prediction from Fabric360 metrics, enabling the AI assistant to operate in reactive (millisecond-latency) rather than scheduled (hourly/daily) mode.
- **Multi-Region Federated Learning:** For privacy-sensitive production deployments where regional data cannot be centralized, federated learning approaches could train CHF and SDP models collaboratively across NA, EU, APAC, and LATAM without pooling raw telemetry data.
- **Reinforcement Learning for Flow Routing:** Replace the rule-based mitigation policy engine with a reinforcement learning agent trained to optimize fabric-wide congestion metrics through adaptive flow rerouting and QoS parameter adjustment, subject to operator-defined constraints.
- **Expanded SKU Coverage:** Extend the hardware recommendation module to cover next-generation Spectrum-X ASICs, BlueField DPU generations, and 800GbE upgrade paths as NVIDIA's product roadmap evolves.

References

- California Science and Technology University (CSTU). (2026). *CSE599 Capstone Project Guidelines*. Emerging Technologies Program. California Science and Technology University (CSTU).
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why Tree-Based Models Still Outperform Deep Learning on Tabular Data. *Advances in Neural Information Processing Systems (NeurIPS 2022)*, 35, 507–520.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 30, 3146–3154.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 30, 4765–4774.
- Meta AI Research. (2022–2024). *Data Center Network Design for AI Workloads*. Meta Engineering Blog. Meta Platforms, Inc.
- Mitra, S. (2026a). *Phase 2 — AI-Infrastructure Capstone: Synthetic Data Generation & Validation Report*. CSE599 Capstone Deliverable. California Science and Technology University (CSTU), July 2026.
- Mitra, S. (2026b). *Phase 3 — AI-Infrastructure Capstone: EDA, Visualization & ML Feature Candidates*. CSE599 Capstone Deliverable. California Science and Technology University (CSTU), July 2026.
- MLCommons. (2024). *MLPerf Training Benchmarks v4.0*. MLCommons Association. <https://mlcommons.org/benchmarks/training/>
- NVIDIA Corporation. (2024). *NVIDIA Spectrum-X Advanced Ethernet Networking Platform: Technical Overview*. NVIDIA Data Center Documentation.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- CAIDA (Center for Applied Internet Data Analysis). (2024). *CAIDA Topology and Traffic Datasets*. University of California San Diego. <https://www.caida.org/>
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Appendix A — Dataset Schema

This appendix provides full field-level schema definitions for all three synthetic corpus tables: MDR, Supply Plan, and Task. These schemas are derived from Fabric360 (the PCPS) and FlowQ observability schemas and serve as the data model foundation for the four ML models described in this report.

A.1 MDR Schema (n=1,200 rows)

Table A.1. MDR Schema — Full Field-Level Definition (n = 1,200 rows).

Field	Type	Description	Schema(s)
region_id	STRING	Geography region code (e.g., USWE, EMEA, APAC)	MDR, Supply Plan, Task
dc_id	STRING	Datacenter identifier; FK → datacenter table	MDR, Supply Plan, Task
suite_id	STRING	Suite identifier; FK → suite table	MDR, Supply Plan, Task
service_owner_id	STRING	Service owner business unit; FK → service_owner table	MDR
milestone_type	ENUM	Close Land Break Ground Handover Complete QSP Handover CEA Need-By	MDR
planned_date	DATE	Originally committed milestone target date	MDR
actual_date	DATE	Realized completion date; NULL if milestone in-progress	MDR
delay_days	INTEGER	actual_date – planned_date in calendar days (target: SDP)	MDR
supply_chain_risk	FLOAT	Composite risk score [0,1] for supply disruption likelihood	MDR
change_log_entries	INTEGER	Count of revision events for this milestone	MDR
qsp_phase	ENUM	Planning quarter: Q1	MDR, Supply Plan

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		Q2 Q3 Q4	
region_type	ENUM	Hyperscale-Core Edge AI-Dense Hybrid	MDR, Supply Plan
ai_zone_flag	BOOLEAN	TRUE if suite is designated an AI Zone (GPU-dense buildout)	MDR, Supply Plan, Task

The MDR (Milestone Data Record) table captures the lifecycle of each capacity suite from initiation through production readiness, and is the source schema for both SDP (Supply Delay Predictor) and CLS (Topology Upgrade Classifier).

A.2 Supply Plan Schema (n=800 rows)

Table A.2. Supply Plan Schema — Full Field-Level Definition (n = 800 rows).

Field	Type	Description	Schema
region_id	STRING	Geography region code	Supply Plan
dc_id	STRING	Datacenter identifier	Supply Plan
suite_id	STRING	Suite identifier	Supply Plan
supply_mw	FLOAT	Committed electrical supply in megawatts	Supply Plan
demand_mw	FLOAT	Forecast electrical demand in megawatts	Supply Plan
headroom_pct	FLOAT	$(\text{supply_mw} - \text{demand_mw}) / \text{supply_mw} \times 100$; target variable for CHF model	Supply Plan
qsp_phase	ENUM	Planning quarter: Q1–Q4	Supply Plan
region_type	ENUM	Hyperscale-Core Edge AI-Dense Hybrid	Supply Plan
ai_zone_flag	BOOLEAN	TRUE if suite is GPU-dense AI Zone	Supply Plan
plan_version	ENUM	Current Silver Gold — plan maturity state	Supply Plan

The Supply Plan table is a quarterly time series of power capacity projections by geographic region, and is the source schema for CHF (Capacity Headroom Forecaster).

A.3 Task Schema (n=600 rows)

Table A.3. Task Schema — Full Field-Level Definition (n = 600 rows).

Field	Type	Description	Schema
region_id	STRING	Geography region code	Task
suite_id	STRING	Suite identifier	Task
service_owner_id	STRING	Service owner business unit	Task
task_type	ENUM	New Build Retrofit NSI AI Zone Retrofit	Task
effort_eng_wks	FLOAT	Engineering effort in engineer-weeks; target for TEE model	Task
task_id	STRING	Unique task record identifier; FK → planning system task	Task
priority	ENUM	Po P1 P2 — task priority tier	Task
planning_bucket	ENUM	New Build Retrofit NSI AI-Zone	Task
topology_upgrade	BOOLEAN	TRUE if fabric reconfiguration required; target for CLS	Task
pto_fraction	FLOAT	PTO deduction as fraction of gross engineer-weeks	Task
oncall_fraction	FLOAT	On-call burden fraction	Task
psc_fraction	FLOAT	People Success Commitment hours fraction	Task
ai_zone_flag	BOOLEAN	TRUE if task involves GPU-dense AI Zone suite	Task

The Task table records discrete engineering and operational tasks associated with capacity lifecycle activities, and is the source schema for TEE (Task Effort Estimator).

— A-1 —

Appendix B — Glossary of Terms

The following terms and abbreviations are used throughout this report. Terms are listed alphabetically.

Adaptive Routing Dynamic per-packet path selection across equal-cost paths to distribute load and avoid congestion in AI cluster fabrics.

All-Reduce Collective communication primitive used in distributed AI training; aggregates gradients across all GPU endpoints simultaneously.

CLOS Multi-tier spine-leaf switching topology providing non-blocking bandwidth across all endpoint pairs. Used in hyperscale AI cluster fabrics.

CLS Topology Upgrade Classifier — binary classification model predicting whether a suite requires an OCS or Expander fabric upgrade.

CHF Capacity Headroom Forecaster — regression model predicting suite-level electrical power headroom as a percentage of committed supply.

DCQCN Data Center Quantized Congestion Notification — congestion control protocol for RoCEv2 transport; manages ECN marking and rate reduction.

ETL Extract, Transform, Load — the three-stage data pipeline process from raw source systems to analytical data stores.

Expander Fabric upgrade option adding port-density expansion modules to existing switches to increase GPU pod attachment capacity.

Fabric360 Synthetic per-port, per-link telemetry plane for AI cluster fabric observability. Provides the primary telemetry input to all four ML models.

FlowQ Synthetic topology- and event-level fabric monitoring system providing flow-level telemetry complement to Fabric360.

GBR Gradient Boosted Regressor (scikit-learn implementation). One of four ensemble models evaluated across regression tasks.

GPU Graphics Processing Unit — primary compute accelerator for distributed AI training workloads. Installed in Scorpii AI Training Racks.

Scorpii AI Training Rack High-density GPU rack unit optimized for distributed AI training. A standardized 72-GPU rack form factor used in AI Zone suite deployments.

HGB HistGradientBoosting — scikit-learn histogram-based gradient boosting algorithm. Fastest training time among evaluated models.

HIVE Apache Hive — distributed analytical data warehouse. Used as the persistent storage layer for ETL-transformed planning and telemetry data.

LGB LightGBM — gradient boosting framework developed by Microsoft. Provides leaf-wise tree growth for higher accuracy on tabular data.

MDR Milestone Data Record — synthetic schema tracking the full lifecycle of datacenter capacity suite delivery milestones from plan to actual completion.

NCCL NVIDIA Collective Communications Library — communication library for all-reduce, broadcast, and other collective operations in distributed GPU training.

NSI Non-Standard Infrastructure — suite-level power supply planning category for deployments outside standard QSP/QTP build and retrofit workflows.

OCS Optical Circuit Switch — fabric upgrade option providing reconfigurable optical interconnect between spine-layer switches for AI Zone buildouts.

PCPS Production Capacity Planning System — the production infrastructure planning and delivery tracking platform whose schema serves as the authoritative reference for this project's synthetic corpus.

POR Plan of Record — the committed milestone sequence for a datacenter construction project (e.g., Close Land → Break Ground → Handover Complete).

QSP Quarterly Supply Planning — the planning cycle managing new datacenter construction from land acquisition through electrical handover.

QTP Quarterly Tactical Planning — the planning cycle managing suite-level retrofit, decommission, and power rebalancing within existing datacenters.

RoCEv2 RDMA over Converged Ethernet version 2 — lossless transport protocol for AI cluster east-west traffic. Requires Priority Flow Control (PFC) and ECN.

RTR Region Type Roadmap — long-range regional capacity planning module tracking planned datacenter additions and retirements across multi-year horizons.

SDP Supply Delay Predictor — regression model predicting supply chain milestone slippage (actual minus planned date) in calendar days.

SHAP SHapley Additive exPlanations — model interpretability framework (Lundberg & Lee, 2017) quantifying each feature's marginal contribution to a prediction.

TEE Task Effort Estimator — regression model predicting engineering task effort in engineer-weeks, accounting for PTO, on-call, and PSC deductions.

XGB XGBoost — gradient boosting framework (Chen & Guestrin, 2016). Best overall performer across SDP and CHF regression tasks in this project.